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(54) **SEMICONDUCTOR STRUCTURE HAVING METAL CONTACT FEATURES**

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H01L 21/285 (2006.01)
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H10D 62/00 (2025.01)
H10D 62/13 (2025.01)
H10D 64/62 (2025.01)

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(58) **Field of Classification Search**

CPC **H01L 21/28518**; **H01L 21/02063**; **H01L 21/02244**; **H01L 29/0847**; **H01L 29/45**; **H01L 29/66636**; **H01L 29/66795**; **H01L 29/7848**; **H01L 29/7851**
USPC **257/384**
See application file for complete search history.

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Primary Examiner — Ori Nadav

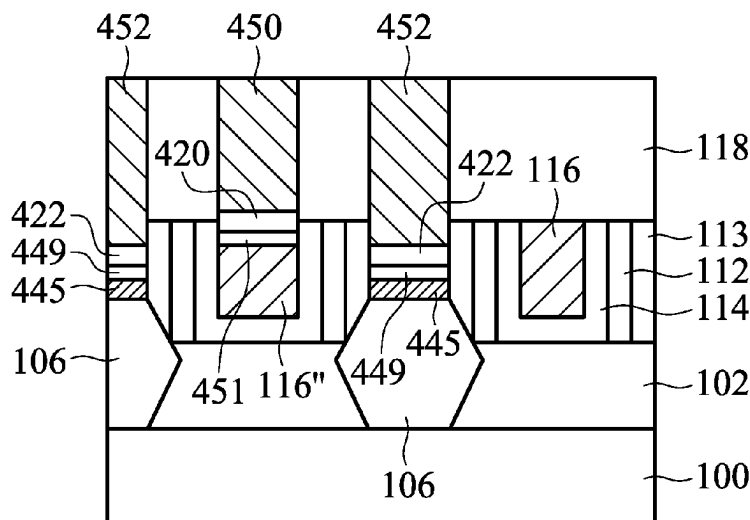
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(57)

ABSTRACT

A semiconductor structure is provided. The semiconductor structure includes an epitaxial structure over a semiconductor substrate. The semiconductor structure also includes a conductive feature over the semiconductor substrate. The conductive feature includes a high-k dielectric layer and a metal layer on the high-k dielectric layer, and a top surface of the metal layer is below a top surface of the high-k dielectric layer. The semiconductor structure further includes a metal-semiconductor compound layer formed on the epitaxial structure. In addition, the semiconductor structure includes a first metal contact structure formed on the top surface of the metal layer of the conductive feature. The semiconductor structure further includes a second metal contact structure formed on the metal-semiconductor compound layer.

20 Claims, 19 Drawing Sheets



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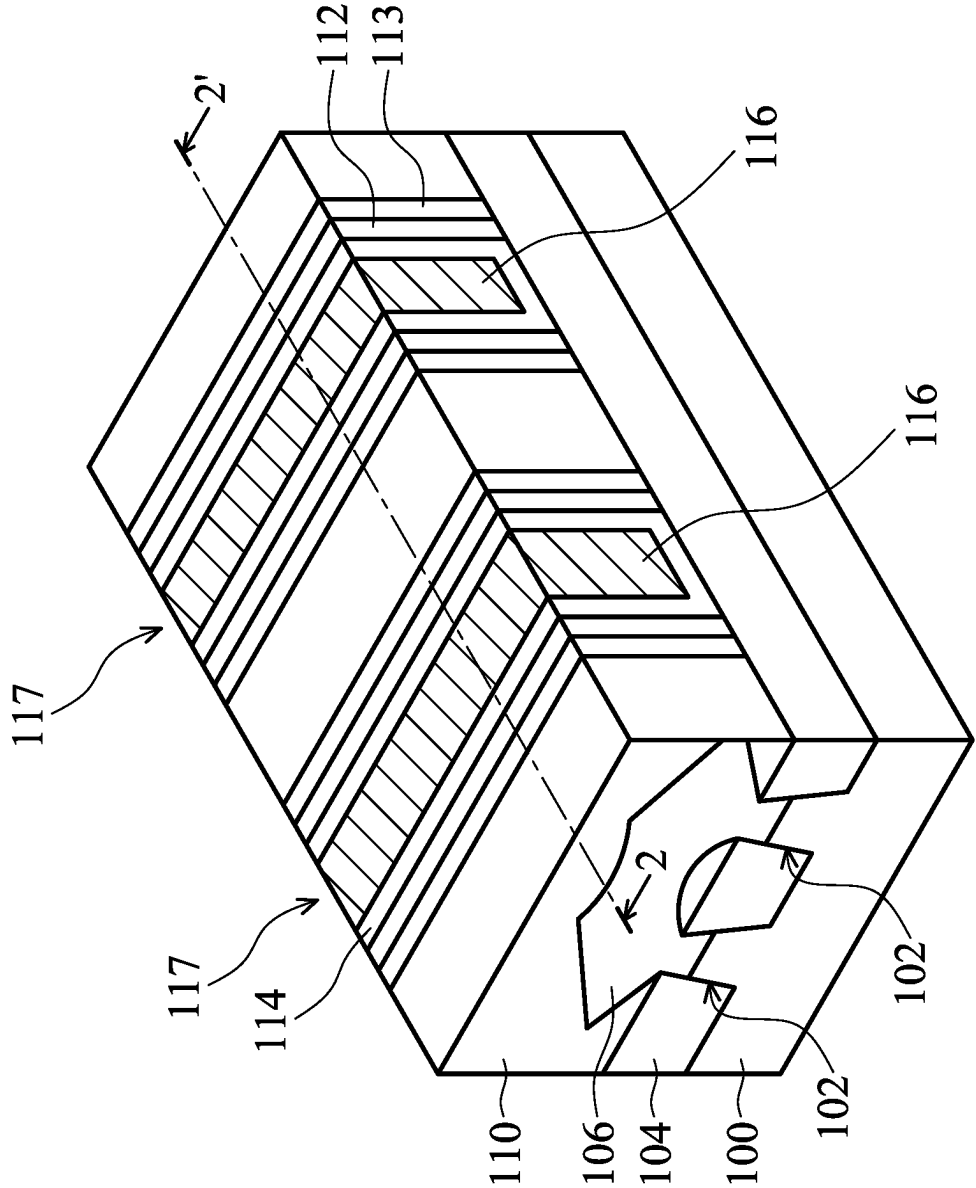


FIG. 1A

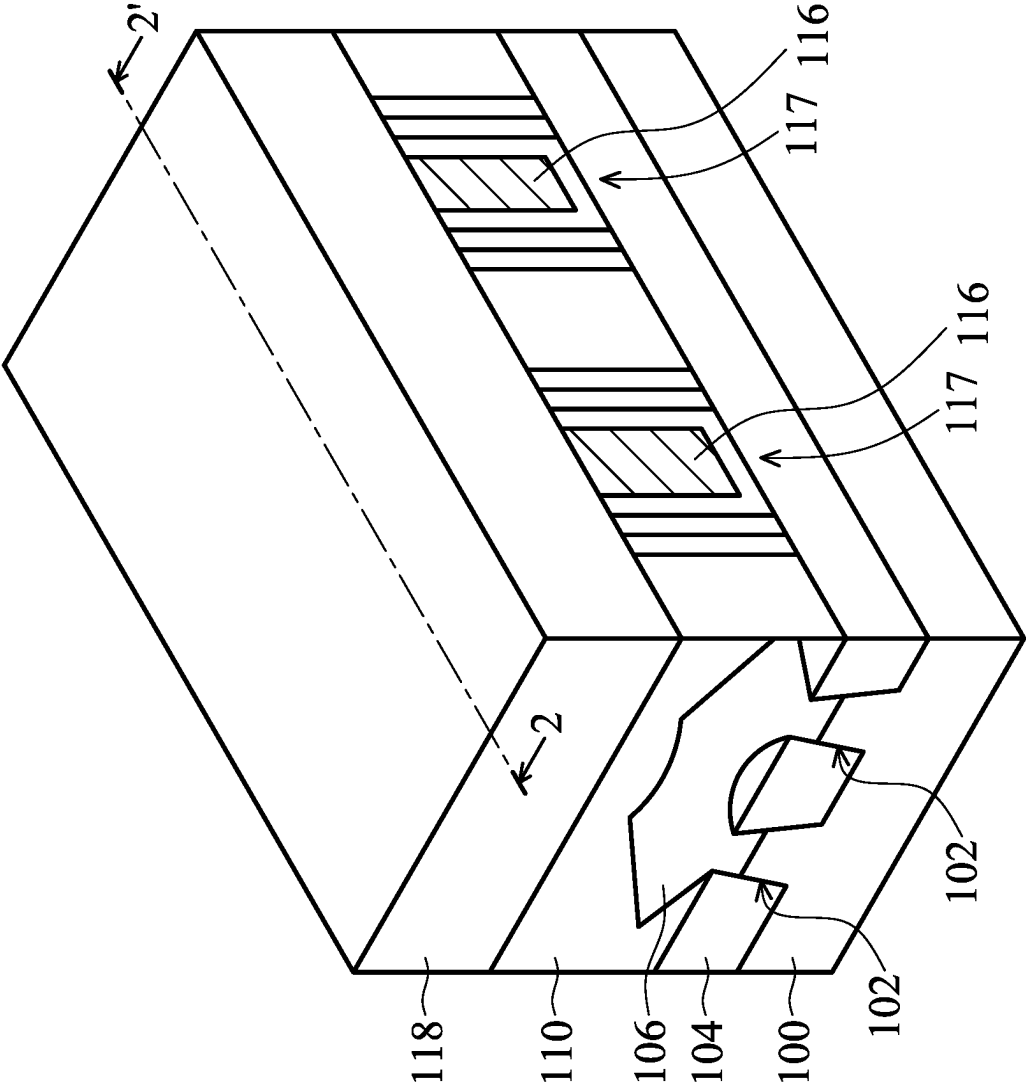


FIG. 1B

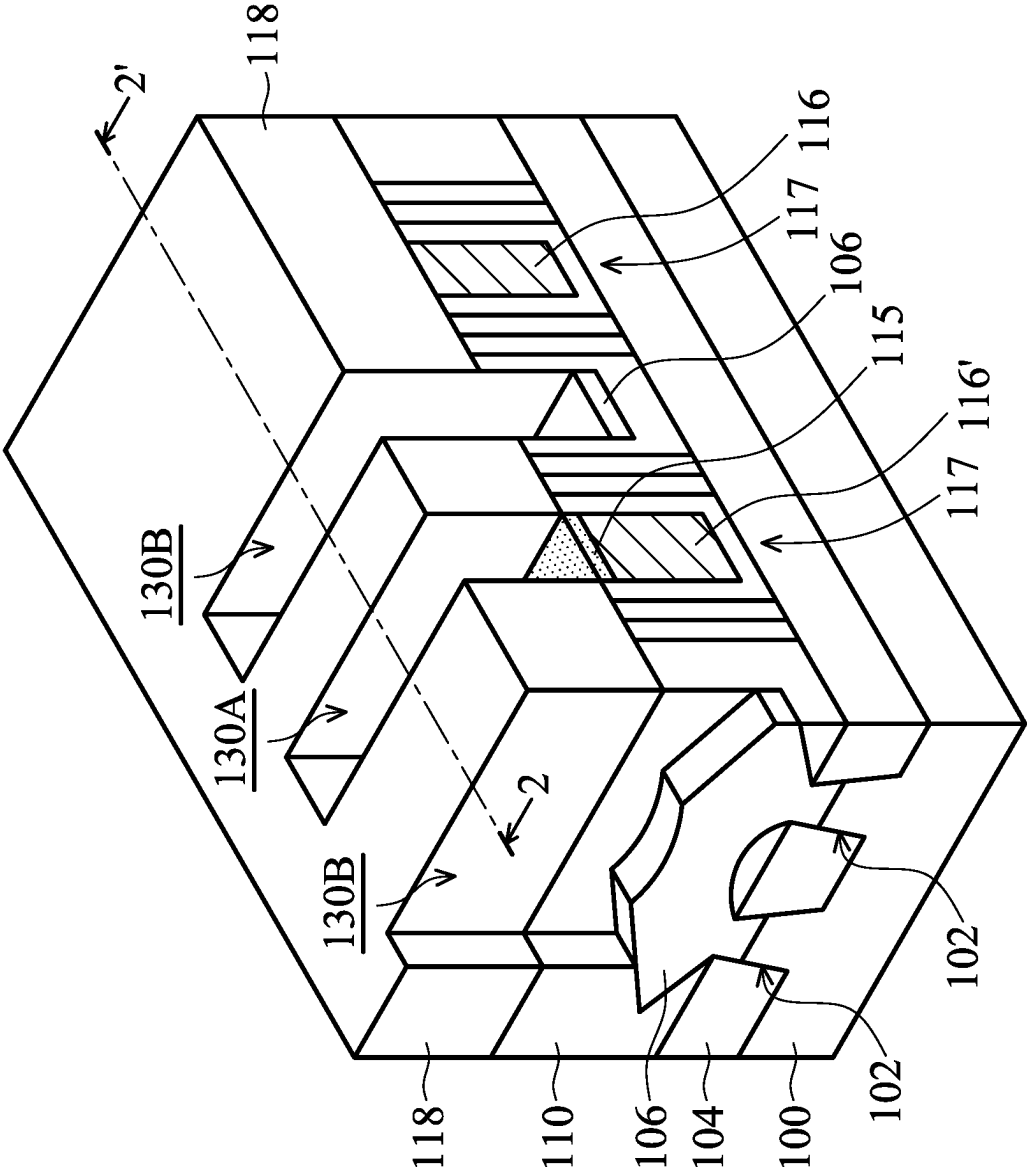


FIG. 1C

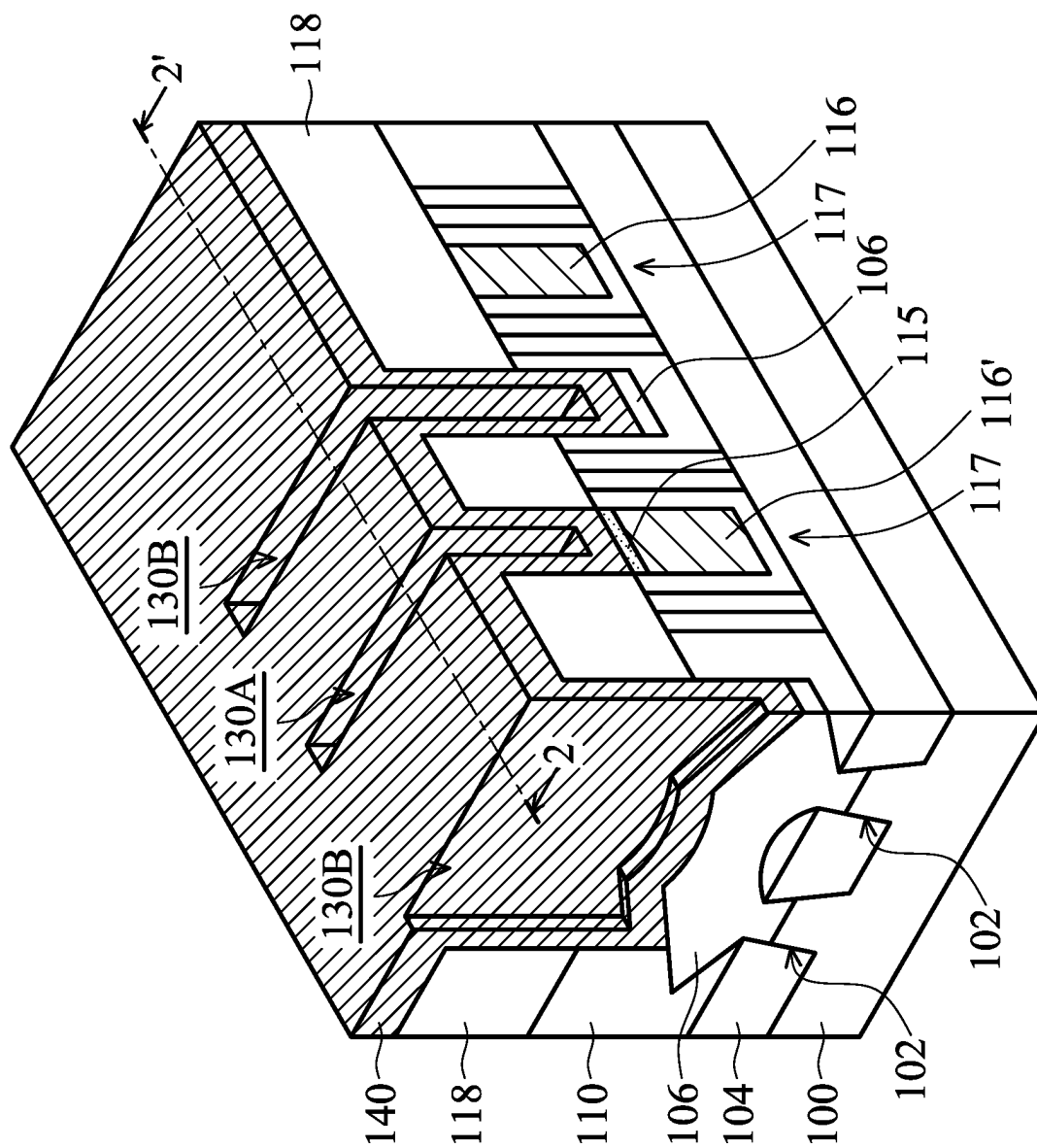


FIG. 1D

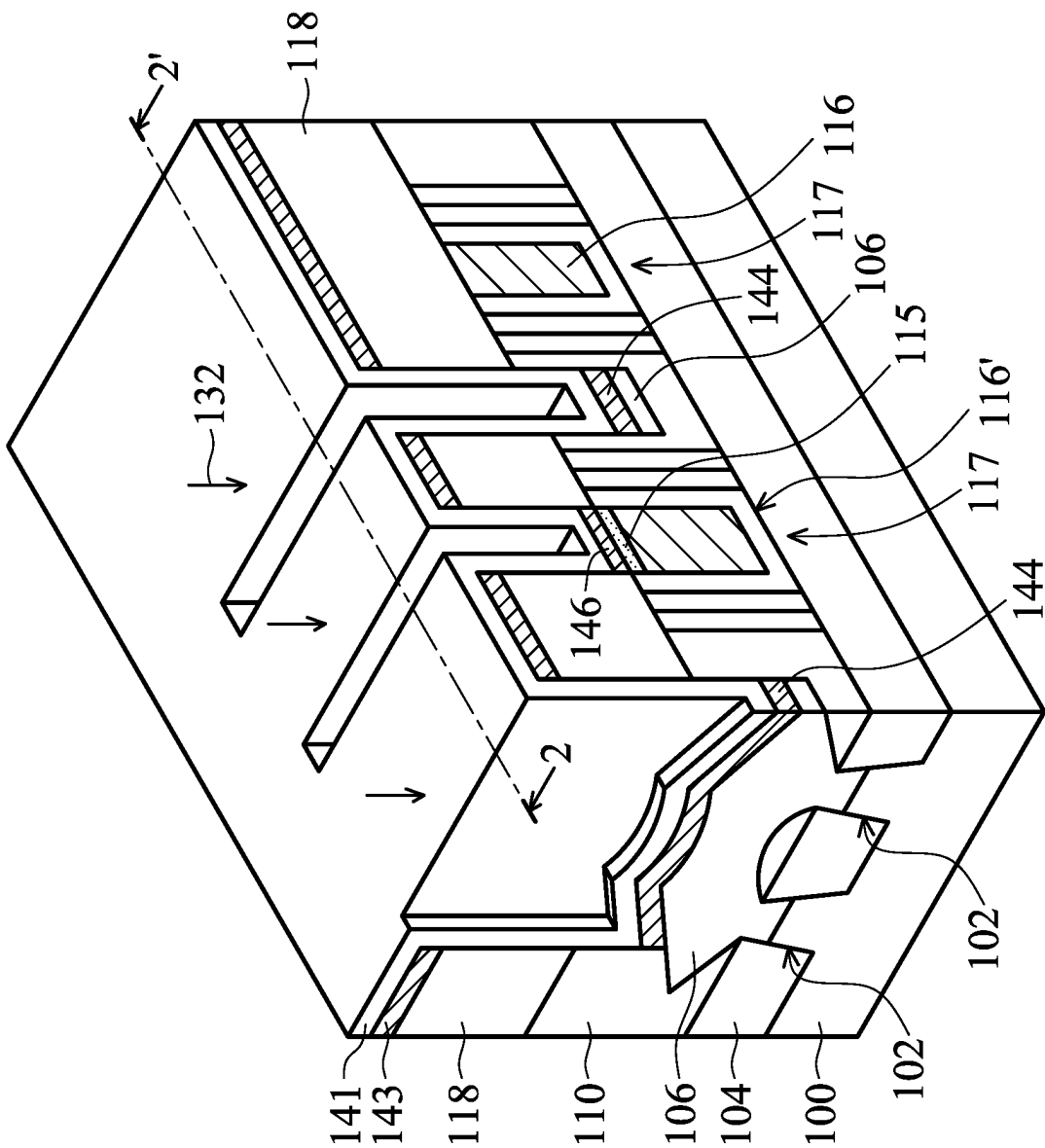


FIG. 1E

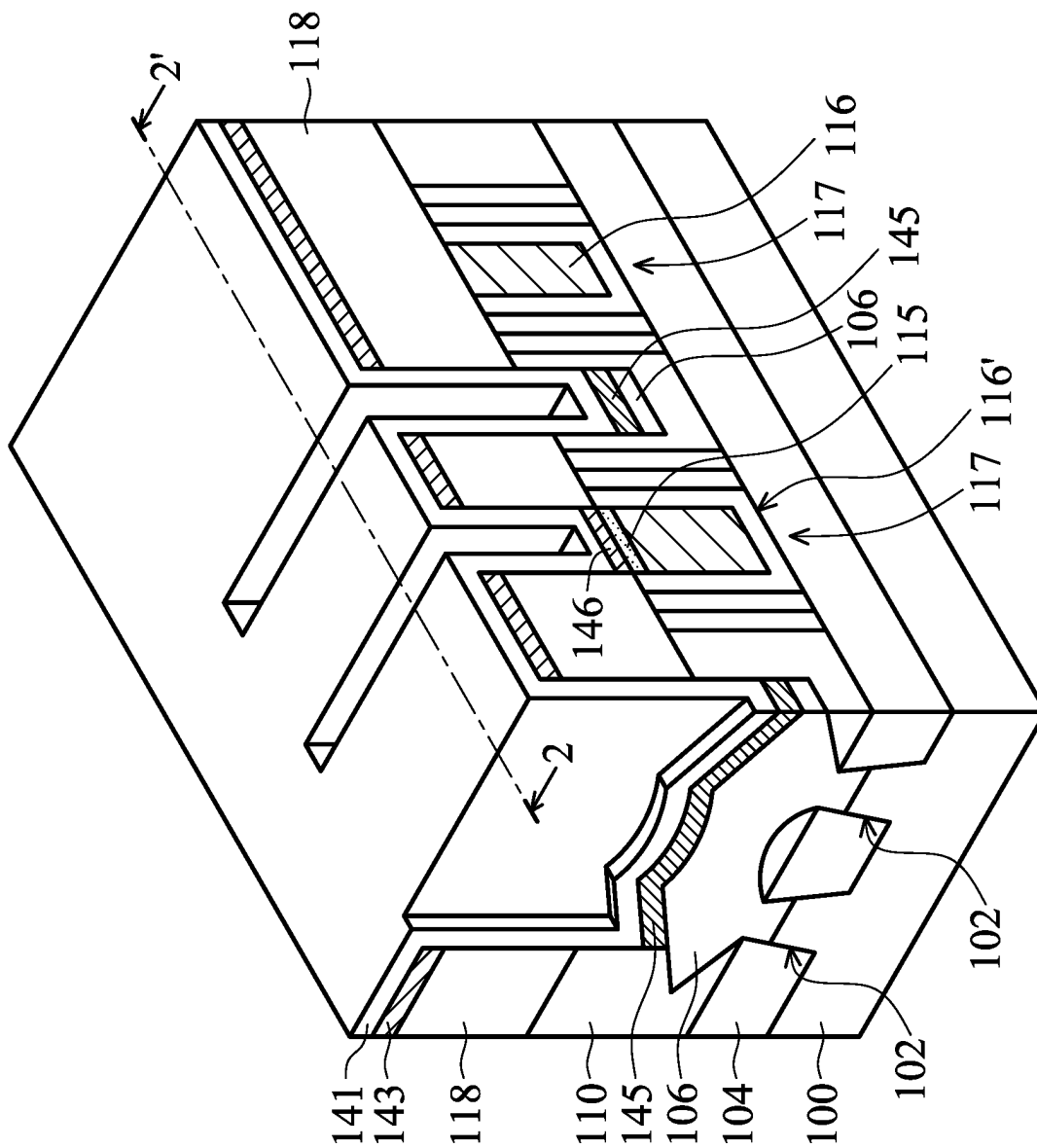


FIG. 1F

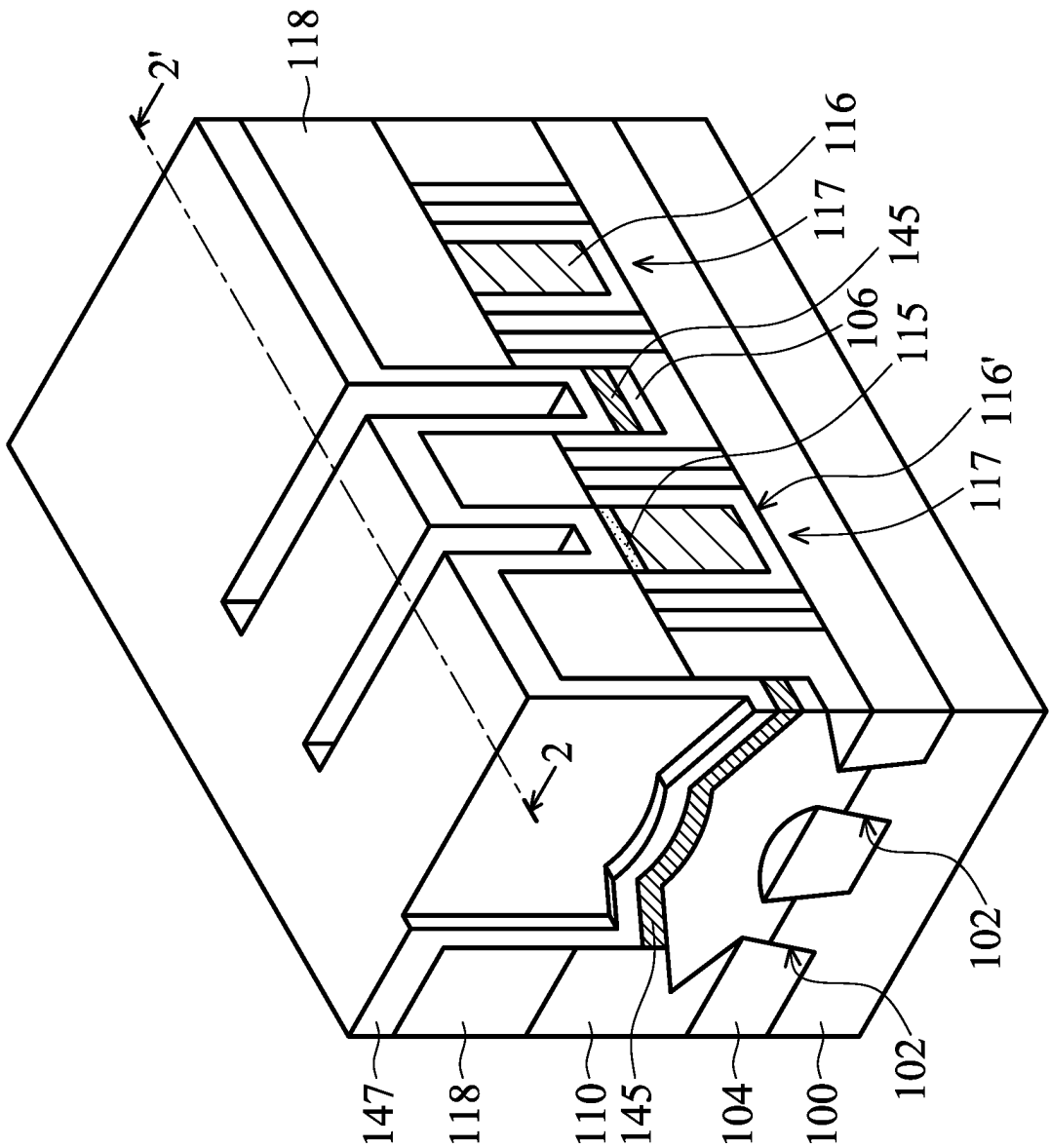


FIG. 1G

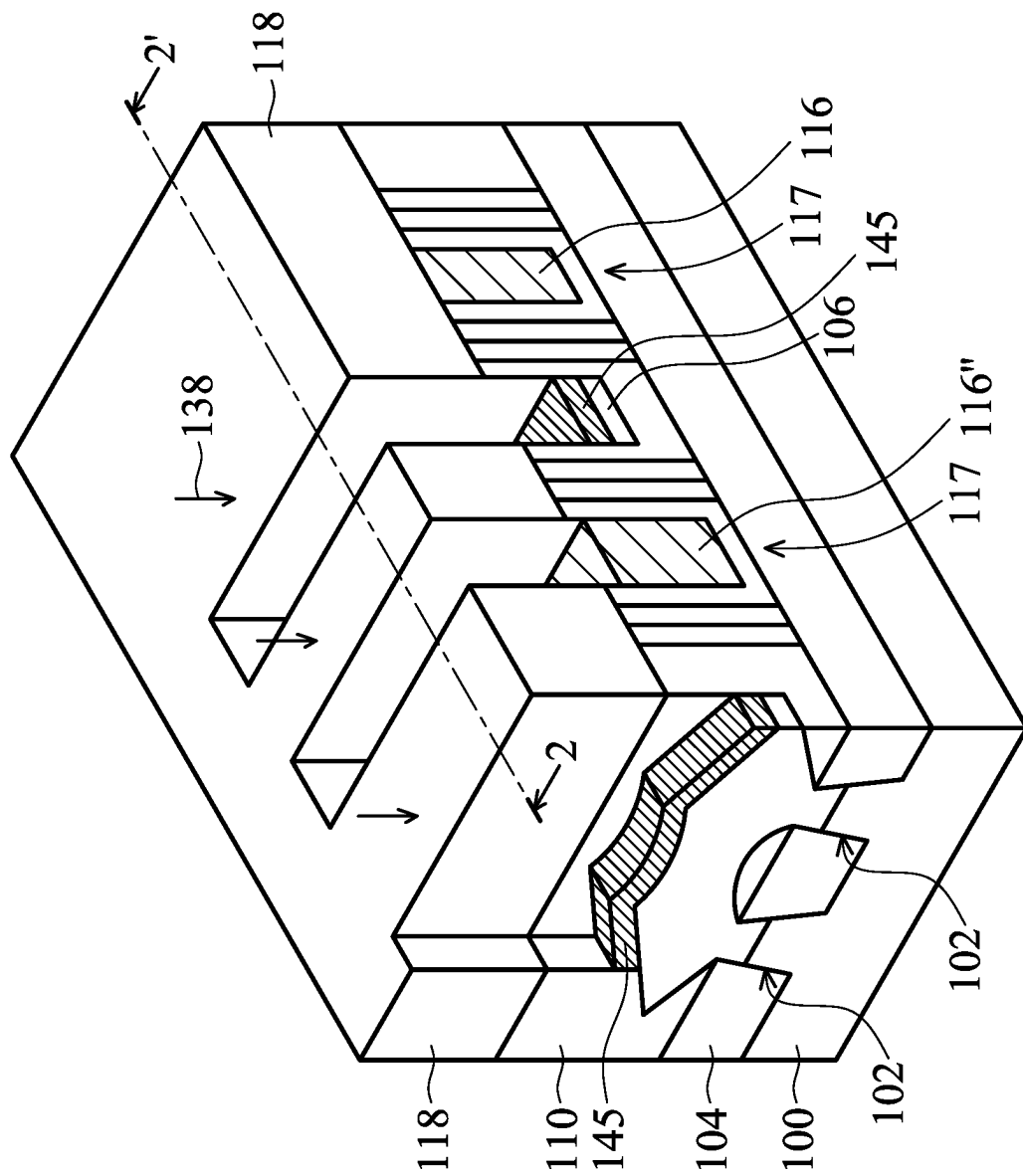


FIG. 1H

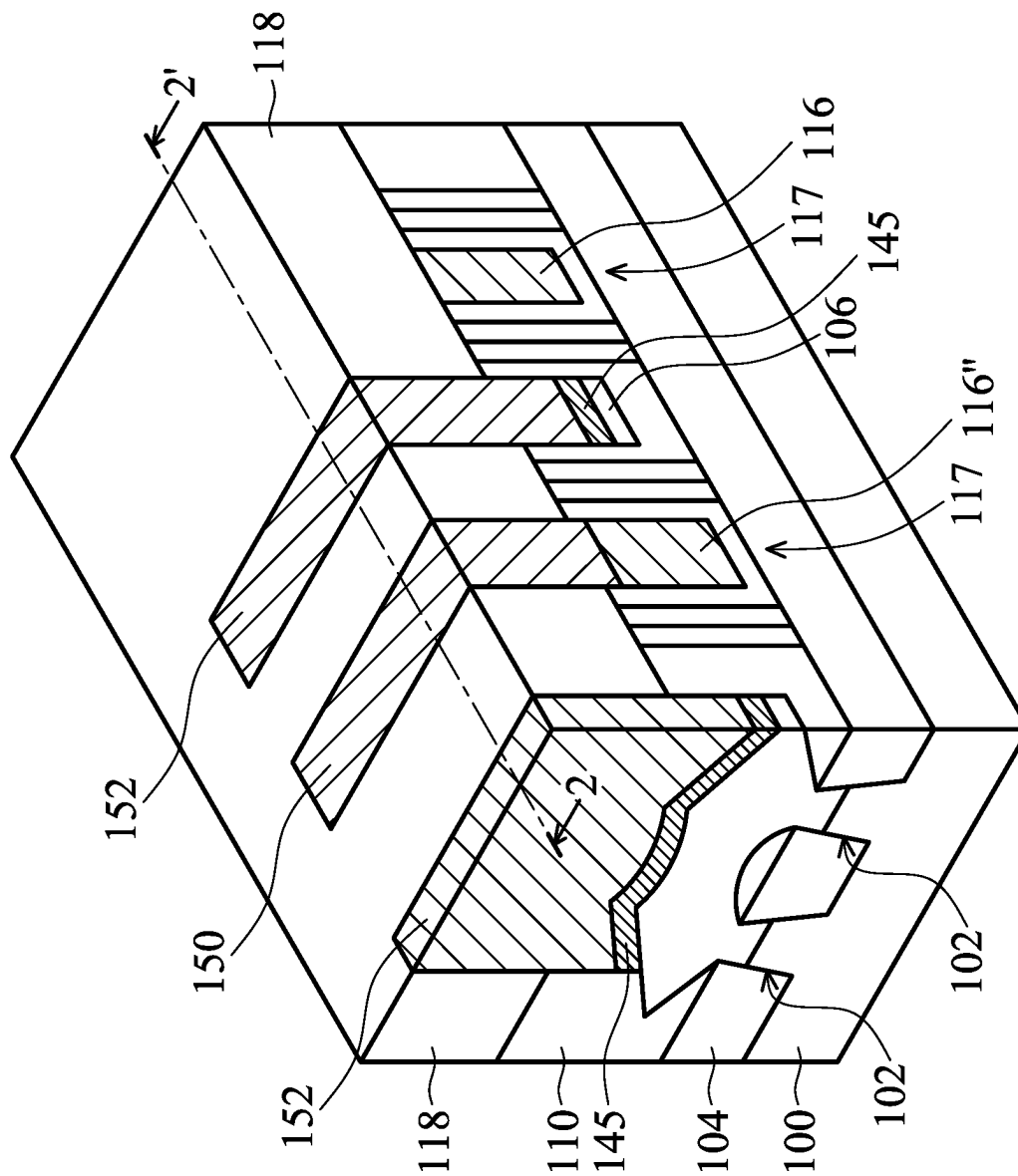


FIG. 11

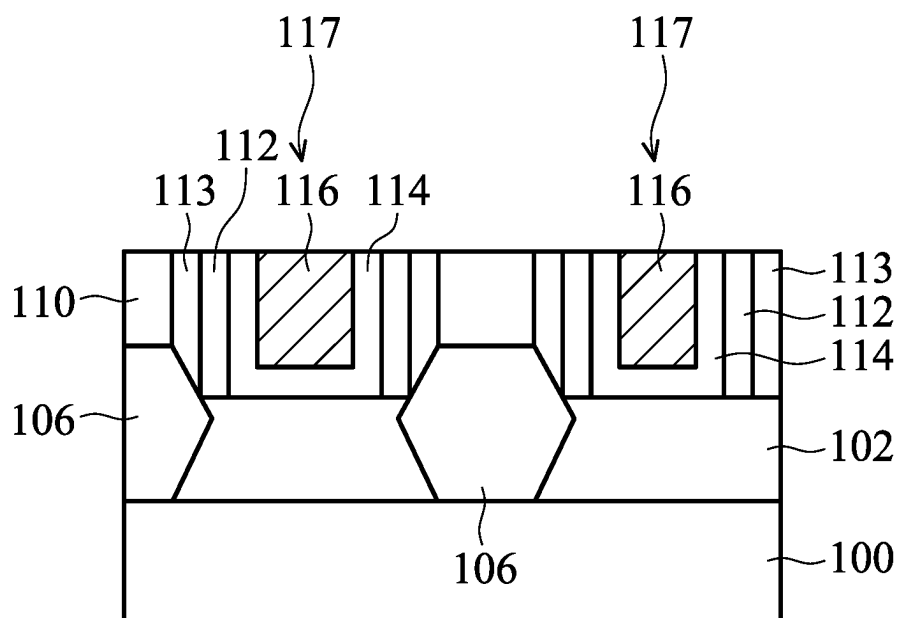


FIG. 2A

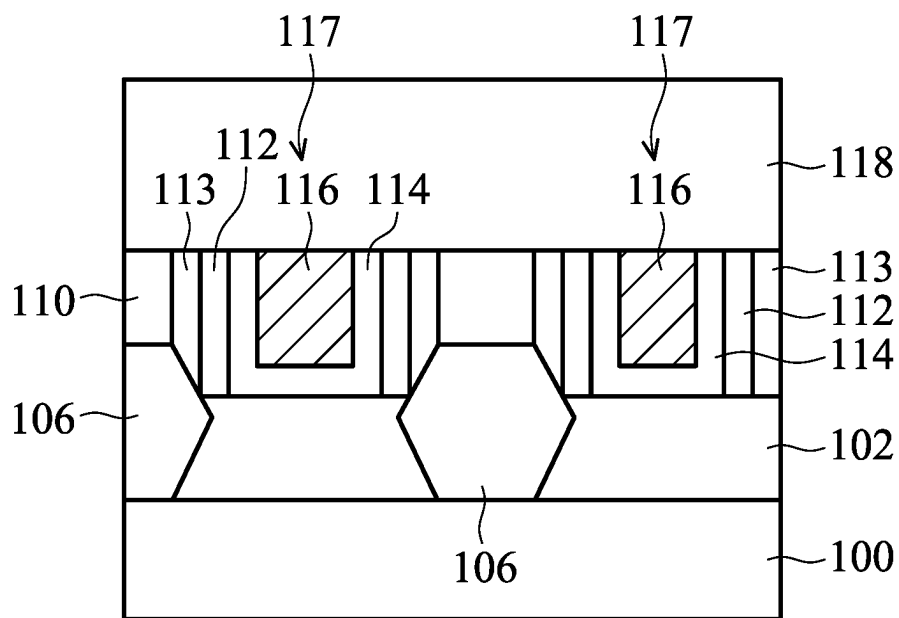


FIG. 2B

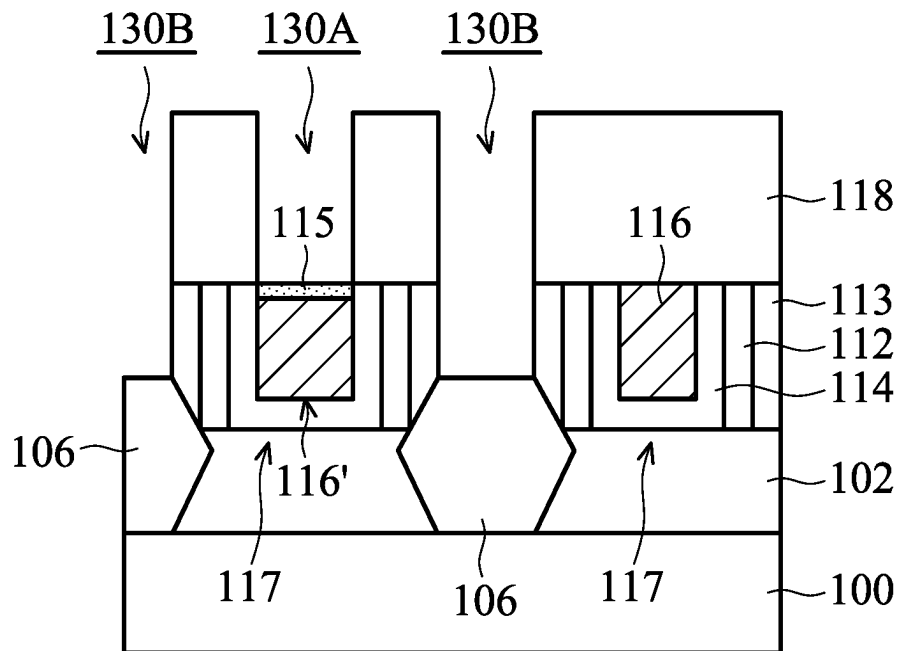


FIG. 2C

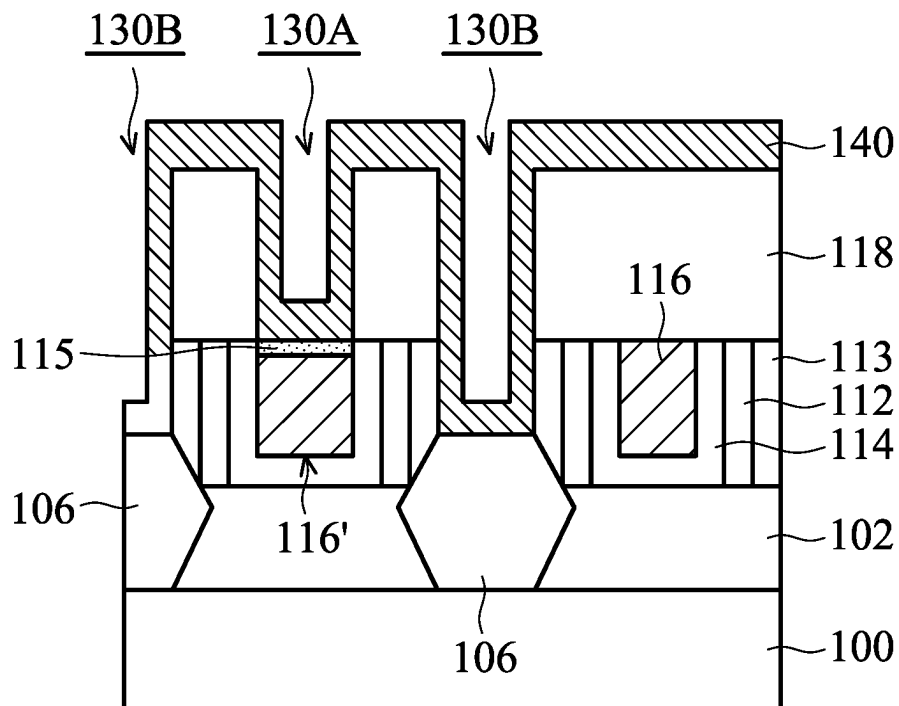


FIG. 2D

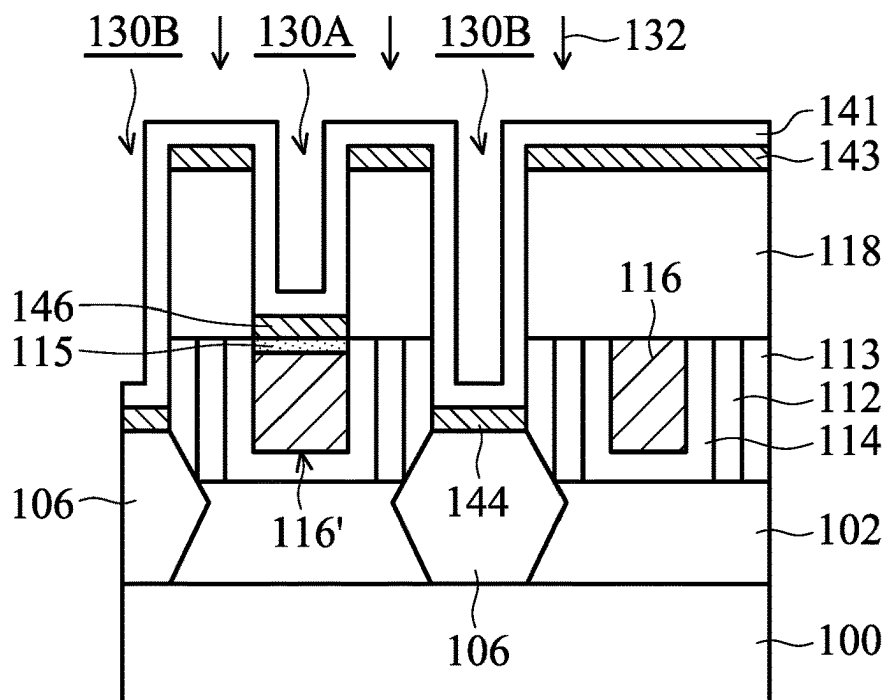


FIG. 2E

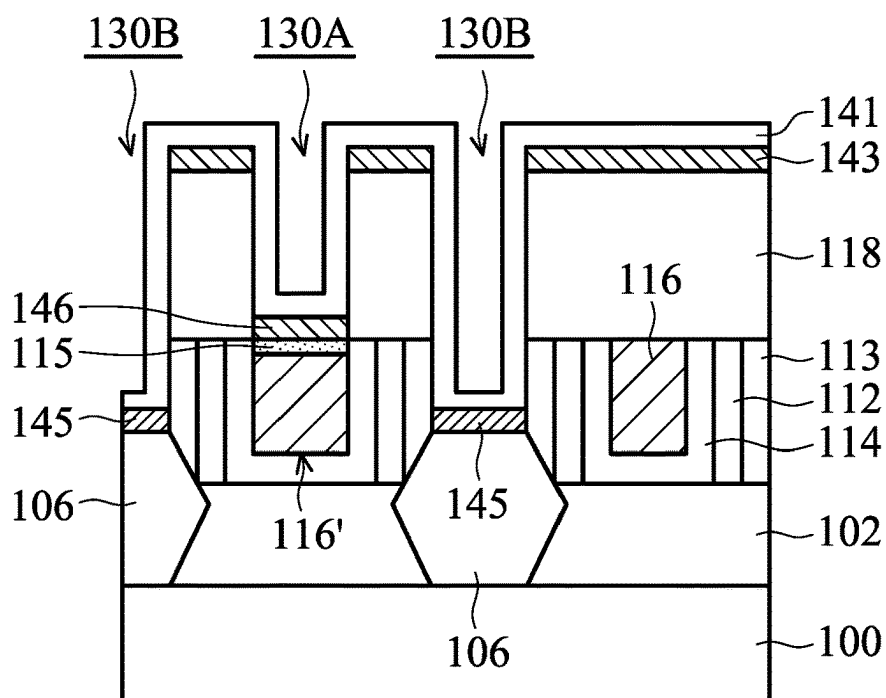


FIG. 2F

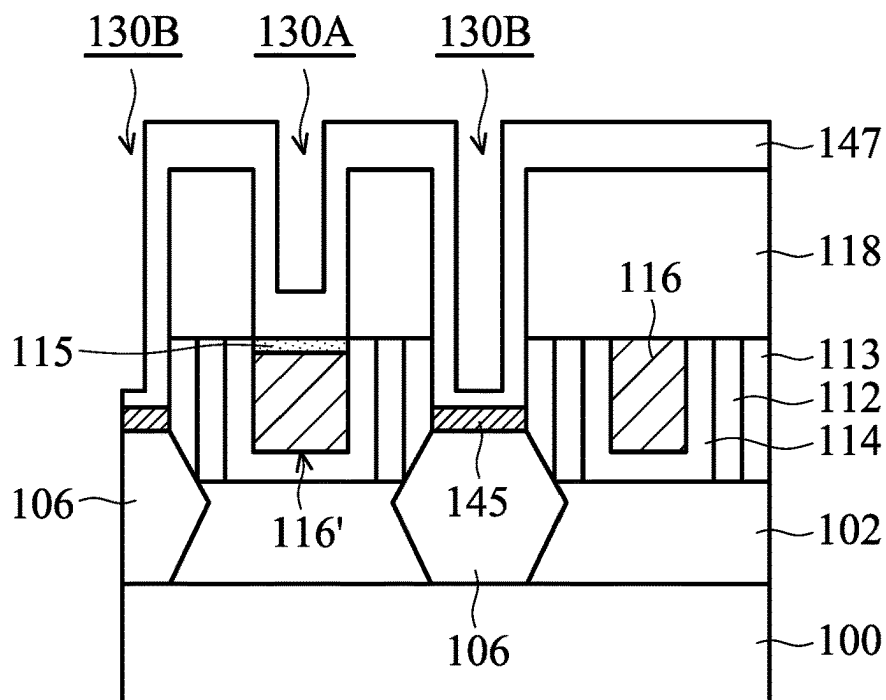


FIG. 2G

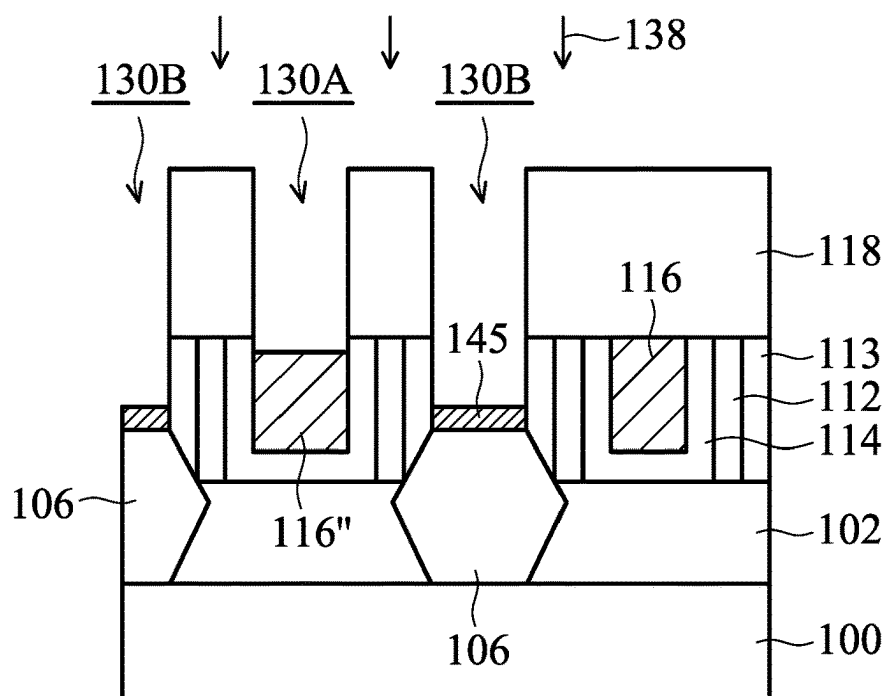


FIG. 2H

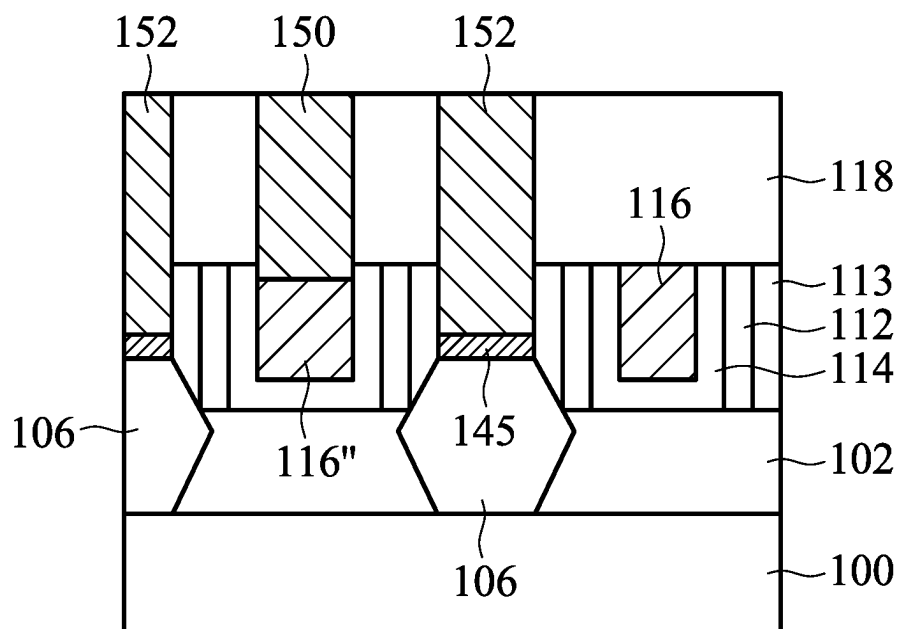


FIG. 2I

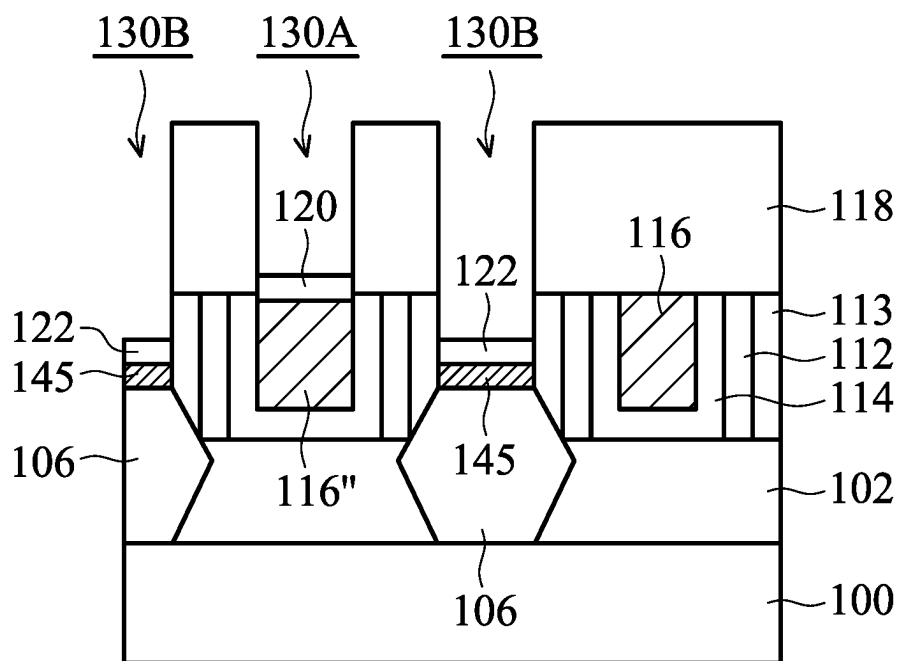


FIG. 3A

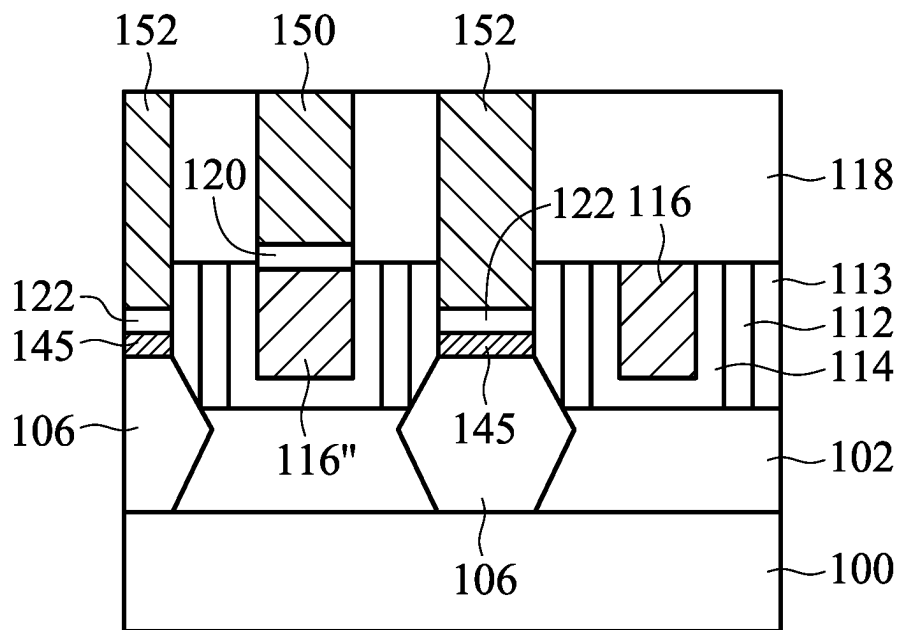


FIG. 3B

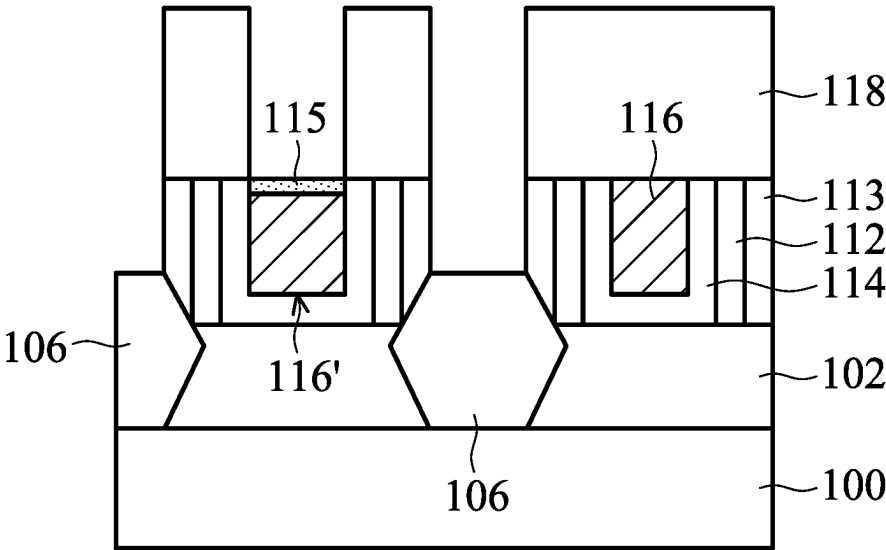


FIG. 4A

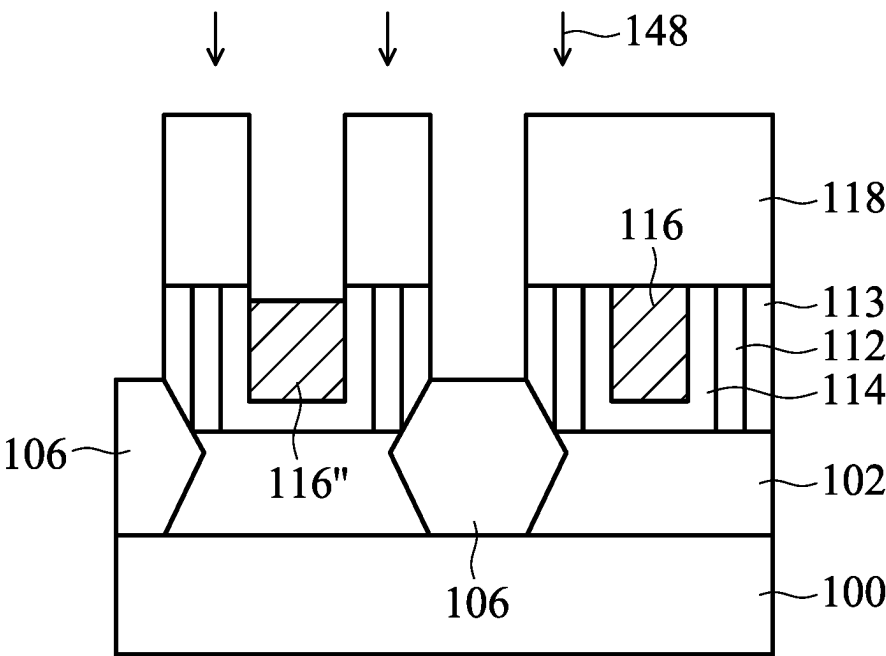


FIG. 4B

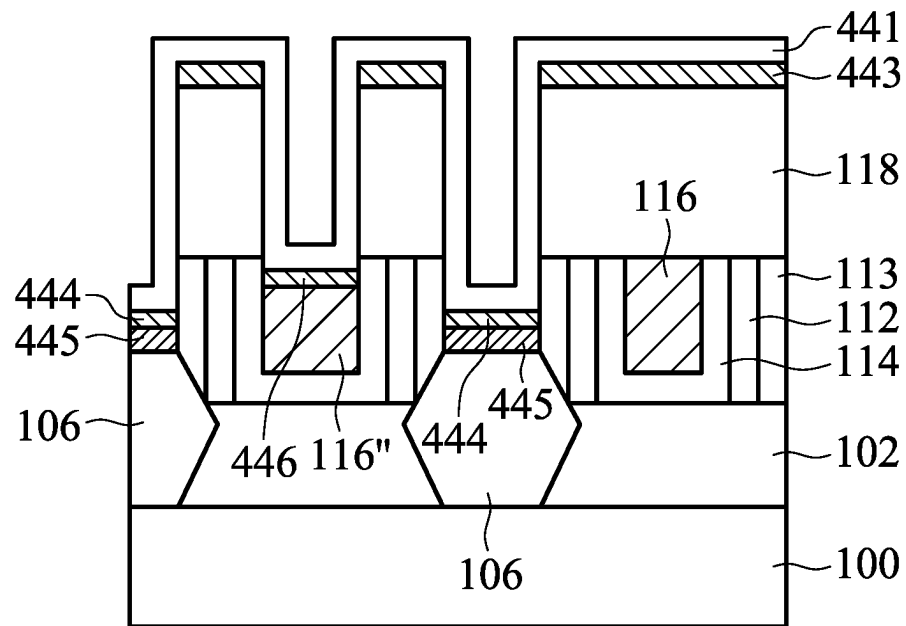


FIG. 4C

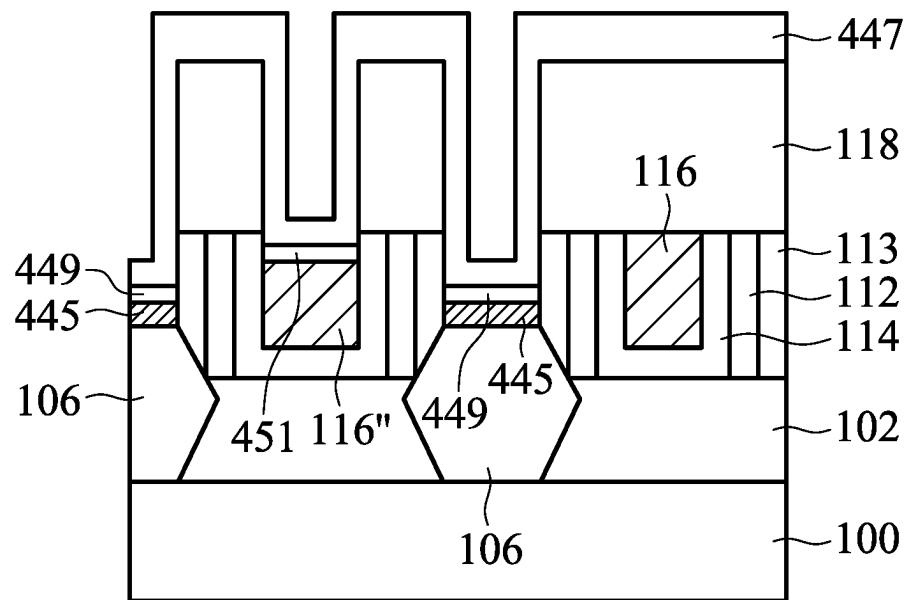


FIG. 4D

FIG. 5

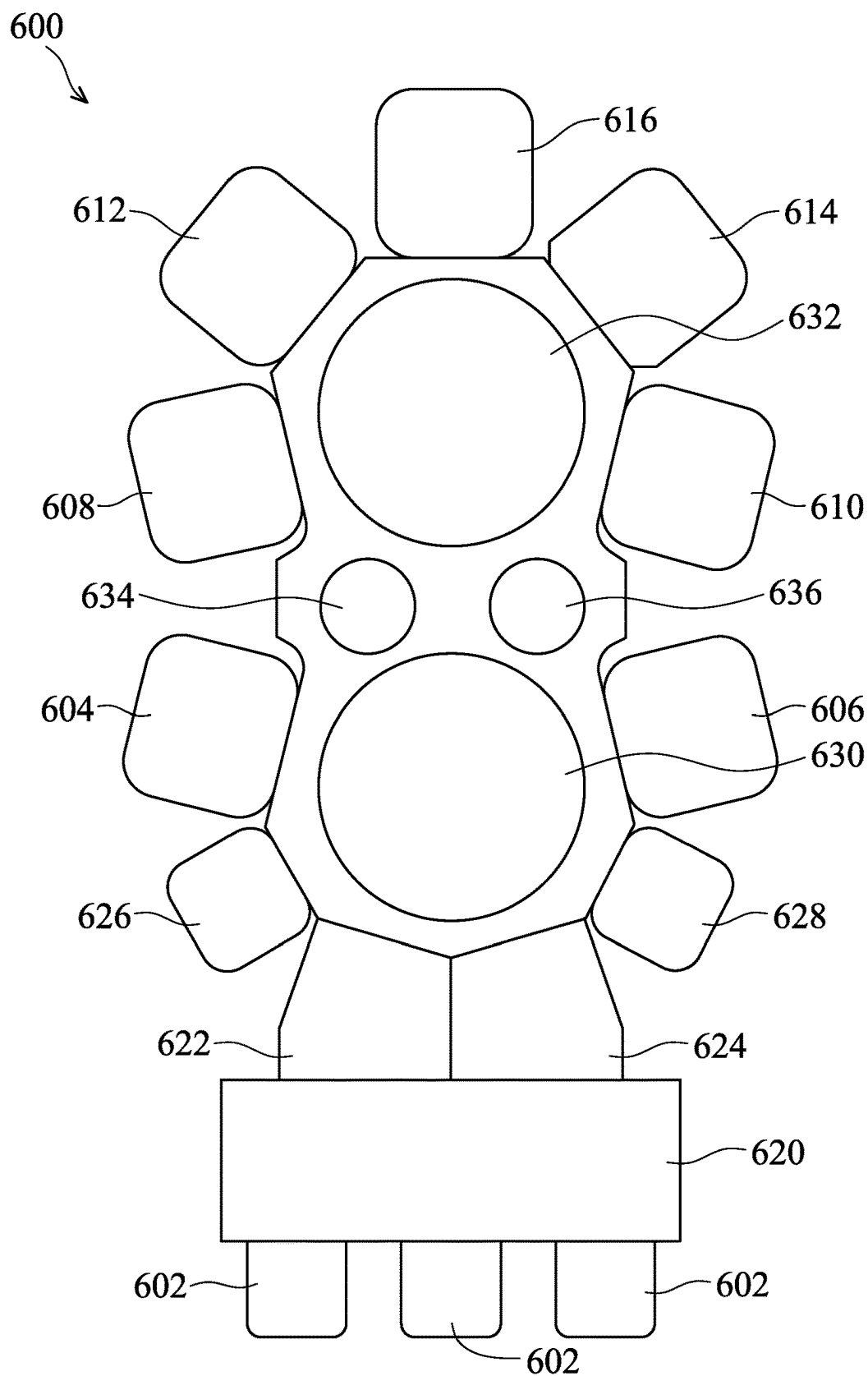


FIG. 6

SEMICONDUCTOR STRUCTURE HAVING METAL CONTACT FEATURES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Division of pending U.S. patent application Ser. No. 16/353,531, filed Mar. 14, 2019 and entitled “SEMICONDUCTOR STRUCTURE HAVING METAL CONTACT FEATURES AND METHOD FOR FORMING THE SAME”, the entirety of which is incorporated by reference herein.

BACKGROUND

The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs.

In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs.

However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIGS. 1A-1I are perspective views of various stages of a process for forming a semiconductor structure, in accordance with some embodiments of the disclosure.

FIGS. 2A-2I are cross-sectional views of various stages of the process for forming the semiconductor structure as shown in FIGS. 1A-1I, in accordance with some embodiments of the disclosure.

FIGS. 3A-3B are cross-sectional views of various stages of a process for forming a semiconductor structure, in accordance with some embodiments of the disclosure.

FIGS. 4A-4E are cross-sectional views of various stages of a process for forming a semiconductor structure, in accordance with some embodiments of the disclosure.

FIG. 5 is cross-sectional views of a semiconductor structure, in accordance with some embodiments of the disclosure.

FIG. 6 shows an in-situ cluster tool used in a process for forming a semiconductor structure, in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different fea-

tures of the subject matter provided. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Some variations of the embodiments are described. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. It should be understood that additional operations can be provided before, during, and/or after the stages described in these embodiments. Some of the stages that are described can be replaced or eliminated for different embodiments. Additional features can be added to the semiconductor device structure. Some of the features described below can be replaced or eliminated for different embodiments. Although some embodiments are discussed with operations performed in a particular order, these operations may be performed in another logical order.

Furthermore, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It should be understood that additional operations can be provided before, during, and after the method, and some of the operations described can be replaced or eliminated for other embodiments of the method.

Embodiments of the disclosure may relate to a FinFET structure having fins. The fins may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

Embodiments for forming a semiconductor structure are provided. The method for forming the semiconductor structure may include transforming a portion of a metal-containing layer over a conductive feature into a metal-semiconductor compound layer, and oxidizing another portion of the metal-containing layer into a metal oxide layer. The metal oxide layer may be removed by using a metal chloride-containing etching gas. In addition, a metal contact feature may be selectively grown on the metal-semiconductor compound layer. Moreover, a metal-containing element that is

substantially fluorine free may be selectively grown on the metal-semiconductor compound layer. The selective growths of the metal-containing element and the metal contact feature may be in-situ performed, thus the electrical connection between the metal contact feature and the conductive feature can be greatly improved.

FIGS. 1A-1I are perspective views of various stages of a process for forming a semiconductor structure **100**, in accordance with some embodiments of the disclosure. FIGS. 2A-2I are cross-sectional views of various stages of the process for forming the semiconductor structure **100** as shown in FIGS. 1A-1I, in accordance with some embodiments of the disclosure. FIGS. 2A-2I are cross-sectional representations taken along cross-sectional lines 2-2' of FIGS. 1A-1I, in accordance with some embodiments of the disclosure.

A semiconductor substrate **100** is provided as shown in FIGS. 1A and 2A, in accordance with some embodiments of the disclosure. In some embodiments, the semiconductor substrate **100** is a bulk semiconductor substrate, such as a semiconductor wafer. For example, the semiconductor substrate **100** includes silicon or other elementary semiconductor materials such as germanium. In some embodiments, the semiconductor substrate **100** may be un-doped or doped (e.g., p-type, n-type, or a combination thereof).

In some other embodiments, the semiconductor substrate **100** includes a compound semiconductor. The compound semiconductor may include silicon carbide, gallium arsenide, indium arsenide, indium phosphide, one or more other applicable compound semiconductors, or a combination thereof.

In some embodiments, the semiconductor substrate **100** is an active layer of a semiconductor-on-insulator (SOI) substrate. The SOI substrate may be fabricated using a separation by implantation of oxygen (SIMOX) process, a wafer bonding process, another applicable method, or a combination thereof. In some embodiments, the semiconductor substrate **100** includes a multi-layered structure. For example, the semiconductor substrate **100** includes a silicon-germanium layer formed on a bulk silicon layer.

Recesses (or trenches) are formed in the semiconductor substrate **100** as shown in FIG. 1A, in accordance with some embodiments. As a result, fin structures are formed or defined between the recesses. Two of the fin structures (the fin structure **102**) are shown in FIG. 1A, and one of the fin structures is shown in FIG. 2A, in accordance with some embodiments. In some embodiments, one or more photolithography and etching processes are used to form the recesses. In some embodiments, the fin structures **102** are in direct contact with the semiconductor substrate **100**.

However, embodiments of the disclosure have many variations and/or modifications. In some other embodiments, the fin structures **102** are not in direct contact with the semiconductor substrate **100**. One or more other material layers may be formed between the semiconductor substrate **100** and the fin structures **102**. For example, a dielectric layer may be formed between the semiconductor substrate **100** and the fin structures **102**.

An isolation feature **104** is formed in the recesses to surround lower portions of the fin structures **102** as shown in FIG. 1A, in accordance with some embodiments. The isolation feature **104** may be used to define and electrically isolate various device elements formed in and/or over the semiconductor substrate **100**. In some embodiments, the isolation feature **104** includes shallow trench isolation (STI)

features, local oxidation of silicon (LOCOS) features, another applicable isolation feature, or a combination thereof.

In some embodiments, the isolation feature **104** has a multi-layer structure. In some embodiments, the isolation feature **104** is made of a dielectric material. The dielectric material may include silicon oxide, silicon nitride, silicon oxynitride, fluoride-doped silicate glass (FSG), low-K dielectric material, one or more other applicable materials, or a combination thereof. In some embodiments, an STI liner (not shown) is formed to reduce crystalline defects at the interface between the semiconductor substrate **100** and the isolation feature **104**. Similarly, the STI liner may also be used to reduce crystalline defects at the interface between the isolation feature **104** and the fin structures **102**.

In some embodiments, a dielectric material layer is deposited over the semiconductor substrate **100**. The dielectric material layer covers the fin structures **102** and fills the recesses between the fin structures **102**. The dielectric material layer may be deposited using a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, a physical vapor deposition (PVD) process, a spin-on process, one or more other applicable processes, or a combination thereof.

In some embodiments, a planarization process is then used to thin down the dielectric material layer until the fin structures **102** or hard mask elements defining the fin structures **102** are exposed. The planarization process may include a chemical mechanical polishing (CMP) process, a grinding process, a dry polishing process, an etching process, one or more other applicable processes, or a combination thereof. Afterwards, the dielectric material layer may be etched back such that the fin structures **102** protrude from the top surface of the remaining dielectric material layer after the etching process. As a result, the remaining portions of the dielectric material layer form the isolation feature **104**, as shown in FIG. 1A, in accordance with some embodiments.

Afterwards, dummy gate stacks (not shown) are formed over the semiconductor substrate **100** to partially cover the fin structures **102**, in accordance with some embodiments. The dummy gate stacks may include a gate electrode and a gate dielectric layer. In some embodiments, a gate dielectric material layer and a gate electrode material layer are deposited over the isolation feature **104** and the fin structures **102**. Afterwards, the gate dielectric material layer and the gate electrode material layer may be patterned to form the dummy gate stacks.

In some embodiments, spacers are formed over sidewalls of the dummy gate stacks. The spacers may be used to assist in the formation of source and drain structures (or regions) in subsequent processes. In some embodiments, each of the spacers includes a first spacer element **112** adjacent to the sidewalls of the corresponding dummy gate stacks and a second spacer element **113** adjacent to the first spacer element **112** as shown in FIGS. 1A and 2A, in accordance with some embodiments. The first spacer element **112** may be used for protecting the dummy gate stacks from damage or loss in subsequent processes. In some embodiments, the first spacer element **112** is made of a low-K material, for example, silicon nitride, silicon oxide, silicon oxynitride, silicon carbide, silicon carbon oxynitride, one or more other applicable materials, or a combination thereof. In some embodiments, the second spacer element **113** is made of a material different from that of the first spacer element **112**, and the second spacer element **113** is made of silicon nitride,

silicon oxide, silicon oxynitride, silicon carbide, silicon carbon oxynitride, one or more other applicable materials, or a combination thereof.

In some embodiments, a first spacer layer and a second spacer layer are deposited over the semiconductor substrate **100**, the fin structures **102**, and the dummy gate stacks. The first spacer layer and the second spacer layer may be deposited using a CVD process, an ALD process, a PVD process, a spin-on process, one or more other applicable processes, or a combination thereof. Afterwards, an etching process, such as an anisotropic etching process, may be performed to partially remove the first spacer layer and the second spacer layer. As a result, the remaining portions of the first spacer layer and the second spacer layer over the sidewalls of the dummy gate stacks form the first spacer element **112** and the second spacer element **113**, respectively.

Afterwards, the fin structures **102** are partially removed to form recesses, in accordance with some embodiments. In some embodiments, the fin structures **102** are recessed to a level below the top surface of the isolation feature **104**. In some embodiments, an etching process is used to form the recesses.

Many variations and/or modifications can be made to embodiments of the disclosure. In some other embodiments, the fin structures **102** are not recessed. The fin structures **102** may be merely thinned without being recessed to a level below the top surface of the isolation feature **104**.

Afterwards, one or more semiconductor materials are epitaxially grown over the fin structures **102** to fill the recesses, in accordance with some embodiments. As a result, epitaxial structures (the epitaxial structure **106**) are formed over the semiconductor substrate **100** as shown in FIGS. 1A and 2A, in accordance with some embodiments. The epitaxial structure **106** may function as a source or drain structure. The epitaxial structure **106** may also function as a stressor to improve carrier mobility. In some embodiments, a merged epitaxial structure **106** is formed on two fin structures **102** as shown in FIG. 1A. In some other embodiments, isolated epitaxial structures are formed on different individual fin structures **102**, respectively (not shown in FIG. 1A).

In some embodiments, the epitaxial structure **106** is formed of a silicon-containing material. In some embodiments, the epitaxial structure **106** is made of one or more p-type semiconductor materials. For example, the epitaxial structure **106** may include epitaxially grown silicon germanium, one or more other applicable epitaxially grown p-type silicon-containing semiconductor materials, or a combination thereof. In some other embodiments, the epitaxial structure **106** is made of one or more n-type semiconductor materials. The epitaxial structure **106** may include epitaxially grown silicon, epitaxially grown silicon carbide (SiC), one or more other applicable epitaxially grown n-type silicon-containing semiconductor materials, or a combination thereof.

In some embodiments, the epitaxial structure **106** is formed using a selective epitaxy growth (SEG) process, a CVD process (e.g., a vapor-phase epitaxy (VPE) process, a low pressure chemical vapor deposition (LPCVD) process, and/or an ultra-high vacuum CVD (UHV-CVD) process), a molecular beam epitaxy process, an ALD process, one or more other applicable processes, or a combination thereof. The formation process of the epitaxial structure **106** may use gaseous and/or liquid precursors.

In some embodiments, the epitaxial structure **106** includes dopants. For example, the epitaxial structure **106** is p-type

doped, and the dopants may include boron, gallium, one or more other applicable dopants, or a combination thereof. For example, the epitaxial structure **106** may also be n-type doped, and the dopants may include phosphor. In some embodiments, the epitaxial structure **106** is doped in-situ during the growth of the epitaxial structure **106**. In some other embodiments, the epitaxial structure **106** is not doped during the growth of the epitaxial structure **106**. In some embodiments, one or more implantation processes are used to dope the epitaxial structure **106**. After the epitaxial growth, the epitaxial structure **106** may be doped in a subsequent process.

In some embodiments, the doping is achieved using an ion implantation process, a plasma immersion ion implantation process, a gas and/or solid source diffusion process, one or more other applicable processes, or a combination thereof. In some embodiments, the epitaxial structure **106** is further exposed to one or more annealing processes to activate the dopants. For example, a rapid thermal annealing process is used.

In some embodiments, a dielectric material layer is deposited over the epitaxial structures **106**, the isolation feature **104**, and the dummy gate stacks. The dielectric material layer may be made of silicon oxide, silicon oxynitride, borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), fluorinated silicate glass (FSG), low-k material, porous dielectric material, one or more other applicable dielectric materials, or a combination thereof. In some embodiments, the dielectric material layer is deposited using a CVD process, an ALD process, a PVD process, a spin-on process, one or more other applicable processes, or a combination thereof.

Afterwards, the dielectric material layer may be thinned until the dummy gate stacks are exposed. The dielectric material layer may be thinned until the hard mask elements used for defining the dummy gate stacks are exposed. Alternatively, the dielectric material layer may be thinned until the dummy gate stacks are exposed. After the thinning process of the dielectric material layer, the remaining portion of the dielectric material layer forms the dielectric layer **110** over the semiconductor substrate **100** as shown in FIGS. 1A and 2A, in accordance with some embodiments. The dielectric layer **110** covers the epitaxial structures **106** and surrounds the dummy gate stacks.

Afterwards, a gate replacement process may be performed. The dummy gate stacks may be removed to form trenches. Then, metal gate stacks **117** may be formed in the trenches. Each of the metal gate stacks **117** may include a high-k gate dielectric layer **114** and a metal layer **116** on the high-k dielectric layer **114**. The metal layer **116** may serve as a metal electrode and include a work function layer and a metal filling surrounded by the work function layer. In some embodiments, the dielectric layer **110** serves as an interlayer dielectric (ILD) and surrounds the metal gate stacks **117**, and a top surface of the dielectric layer **110** is substantially level with a top surface of the metal layer **116** and a top surface of the high-k dielectric layer **114** as shown in FIGS. 1A and 2A.

Afterwards, a dielectric layer **118** is deposited over the dielectric layer **110** and the metal gate stacks **117** as shown in FIGS. 1B and 2B, in accordance with some embodiments. In some embodiments, the dielectric layer **118** is formed so as to cover the metal layers **116** and the epitaxial structures **106**. The material and the formation method of the dielectric layer **118** may be the same as or similar to those of the dielectric layer **110**.

Afterwards, an opening **130A** is formed to expose the metal gate stack **117**, and openings **130B** are formed to expose top surfaces of the epitaxial structures **106**, as shown in FIGS. **1C** and **2C**, in accordance with some embodiments. In some embodiments, the openings **130A** and **130B** are formed in the dielectric layer **110** and the dielectric layer **118** to respectively expose the metal layer **116** and the epitaxial structures **106** as shown in FIGS. **1C** and **2C**. The formation of the openings **130A** and **130B** may involve one or more photolithography processes and one or more etching processes.

In some embodiments, after the metal layer **116** of the metal gate stack **117** is exposed by the opening **130A**, the metal layer **116** is slightly oxidized to form a native oxide layer **115** of the metal layer **116'** as shown in FIGS. **1C** and **2C**. In some embodiments, the native oxide layer **115** has a thickness that is in a range from about 1 nm to about 2 nm. In some embodiments, the native oxide layer **115** is made of an oxide of the metal filling and/or the work function layer of the metal layer **116**.

Afterwards, a metal material **140** is deposited over the dielectric layer **118**, the top surfaces of the epitaxial structures **106**, and the metal layer **116'** as shown in FIGS. **1D** and **2D**, in accordance with some embodiments. In some embodiments, the metal material **140** is deposited directly on sidewalls and bottoms of the openings **130A** and **130B** as shown in FIGS. **1D** and **2D**. In some embodiments, the metal material **140** is made of titanium. In some embodiments, the metal material **140** is made of nickel, cobalt, germanium, one or more other applicable metal materials, or a combination thereof. The metal material **140** may be deposited using a CVD process, an ALD process, a PVD process, one or more other applicable processes, or a combination thereof.

In some embodiments, the metal material **140** formed on the dielectric layer **118** and the bottoms of the openings **130A** and **130B** has a thickness larger than that of the metal material **140** formed on the sidewalls of the openings **130A** and **130B**. In some embodiments, a ratio of the thickness of the metal material **140** formed on the dielectric layer **118** and the thickness of the metal material **140** formed on the sidewalls of the openings **130A** and **130B** is in a range from about 4 to about 6. In some embodiments, the thickness of the metal material **140** formed on the dielectric layer **118** is in a range from about 12 nm to about 13 nm. In some embodiments, the thickness of the metal material **140** formed on the bottoms of the openings **130A** and **130B** is in a range from about 7 nm to about 8 nm. In some embodiments, the thickness of the metal material **140** formed on the sidewalls of the openings **130A** and **130B** is in a range from about 4 nm to about 6 nm.

Afterwards, a nitriding operation **132** is used for nitriding a portion of the metal material **140** to form a metal nitride layer **141** over remaining portions (e.g. remaining portions **143**, **144** and **146**) of the metal material **140** as shown in FIGS. **1E** and **2E**, in accordance with some embodiments. The metal nitride layer **141** may be used for protecting the remaining portions **143**, **144** and **146** of the metal material **140** from being oxidized or contamination, so that resistance between a metal-semiconductor compound layer to be formed from the remaining portion **144** and a metal contact feature to be formed over the epitaxial structure **106** can be reduced. In some embodiments, the metal nitride layer **141**

is made of titanium nitride, nickel nitride, cobalt nitride, germanium nitride, one or more other applicable metal nitride materials, or a combination thereof.

In some embodiments, during the nitriding operation **132**, the metal material **140** is exposed to a nitrogen-containing gas at an elevated temperature. In some embodiments, the nitrogen-containing gas includes ammonia (NH₃), nitrogen gas, one or more other applicable nitrogen-containing gases, or a combination thereof. In some embodiments, the operation temperature may be in a range from about 400 degrees C. to about 550 degrees C. In some embodiments, plasma is applied during the nitriding operation **132** to enhance the adhesion between the as-formed metal nitride layer **141** and the remaining portions **143**, **144** and **146** of the metal material **140**.

In some embodiments, the metal material **140** formed on the sidewalls of the openings **130A** and **130B** are substantially nitrided into the metal nitride layer **141**, as shown in FIGS. **1E** and **2E**. In some embodiments, the remaining portion **143** of the metal material **140** is between the metal nitride layer **141** and the dielectric layer **118**, the remaining portions **144** of the metal material **140** are contacting the epitaxial structures **106**, and the remaining portion **146** of the metal material **140** is between the metal nitride layer **141** and the native oxide layer **115**, as shown in FIGS. **1E** and **2E**. In some embodiments, the metal nitride layer **141** has a thickness that is in a range from about 4 nm to about 6 nm, which may be controlled by the operation time. In some embodiments, the remaining portion **143** of the metal material **140** has a thickness that is in a range from about 7 nm to about 8 nm. In some embodiments, the remaining portion **144** of the metal material **140** has a thickness that is in a range from about 4 nm to about 6 nm. In some embodiments, the remaining portion **146** of the metal material **140** has a thickness that is in a range from about 4 nm to about 6 nm.

Afterwards, the epitaxial structure **106** and the metal-containing layer including the metal nitride layer **141** and the remaining portions **144** of the metal material **140** are heated to transform a first portion of the metal-containing layer contacting the top surfaces of the epitaxial structures **106** into metal-semiconductor compound layers **145** as shown in FIGS. **1F** and **2F**, in accordance with some embodiments. In some embodiments, the first portion of the metal-containing layer is included within the remaining portions **144** of the metal material **140**. The metal-semiconductor compound layer **145** formed over the epitaxial structure **106** can help to reduce resistance between the epitaxial structure **106** and a metal contact feature to be formed over the epitaxial structure **106**. In some embodiments, the metal-semiconductor compound layer **145** is made of a metal silicide material. In some embodiments, the metal-semiconductor compound layer **145** is made of titanium silicide. In some embodiments, the metal-semiconductor compound layer **145** is made of nickel silicide, cobalt silicide, germanium silicide, one or more other applicable metal silicide materials, or a combination thereof. In some embodiments, the metal-semiconductor compound layer **145** has a thickness that is in a range from about 4 nm to about 6 nm.

In some embodiments, a rapid thermal annealing (RTA) process is used to form the metal-semiconductor compound layer **145**. In some embodiments, the operation temperature is in a range from about 500 degrees C. to about 650 degrees C. (e.g. about 575 degrees C.). In some embodiments, the operation time is in a range from about 10 seconds to about 30 seconds (e.g. about 15 seconds).

In some embodiments, a portion of the top surface of the epitaxial structure **106** is modified to have lower crystallinity

than that of an inner portion of the epitaxial structure **106**. The lower crystallinity of the portion of the top surface can facilitate the subsequent formation of the metal-semiconductor compound layer **145**. The portion of the top surface may be modified using an ion implantation process. In some embodiments, because the metal-containing layer is in contact with the top surface of the epitaxial structure **106** during the heating, the thermal energy can help to initiate a chemical reaction between the portion of the top surface of the epitaxial structure **106** and the metal-containing layer. As a result, a portion (e.g. the portion with low crystallinity) of the top surface of the epitaxial structure **106** reacts with the metal-containing layer (e.g. the remaining portion **144** of the metal material **140**) and is transformed into the metal-semiconductor compound layer **145**, in accordance with some embodiments. In some embodiments, after the metal-semiconductor compound layer **145** is formed, the remaining portions **143** and **146** of the metal material **140** formed on the dielectric layer **118** remains untransformed. In some embodiments, the remaining portion **144** of the metal material **140** may not be fully transformed into the metal-semiconductor compound layer **145**, and a residual part (not shown) of the remaining portion **144** of the metal material **140** may remain untransformed between the metal-semiconductor compound layer **145** and the metal nitride layer **141**.

Afterwards, the metal-containing layer is oxidized to transform a second portion of the metal-containing layer over the metal-semiconductor compound layers **145**, the metal layer **116'**, and the dielectric layer **118** into a metal oxide layer **147** as shown in FIGS. **1G** and **2G**, in accordance with some embodiments. In some embodiments, the metal nitride layer **141**, the remaining portions **143** and **146** of the metal material **140**, and optionally the residual part (not shown) of the remaining portion **144** of the metal material **140** on the metal-semiconductor compound layer **145** are oxidized into the metal oxide layer **147**. In some embodiments, the metal oxide layer **147** is made of titanium oxide. In some embodiments, the metal oxide layer **147** is made of nickel oxide, cobalt oxide, germanium oxide, one or more other applicable metal oxide materials, or a combination thereof. In some embodiments, oxidizing the metal-containing layer includes performing a plasma oxidation process, a thermal oxidation process, or a combination thereof on the metal-containing layer.

In some embodiments, a plasma oxidation process is used to form the metal oxide layer **147** as shown in FIGS. **1G** and **2G**. The operation temperature of the plasma oxidation process may be in a range from about 160 degrees C. to about 250 degrees C. The flow rate of oxygen gas may be in a range from about 2000 sccm to about 6000 sccm. Oxygen (O_2) plasma may be applied on the metal-containing layer. The O_2 plasma oxidation may be conducted with a source power that is in a range from about 100 W to about 1500 W and optionally with a bias power that is in a range from about 10 W to about 100 W. In some embodiments, argon gas is also used.

In some embodiments, a thermal oxidation process in air, He/O_2 , Ar/O_2 , or any combination thereof is used to form the metal oxide layer **147** as shown in FIGS. **1G** and **2G**. The operation temperature of the thermal oxidation process may be in a range from about 400 degrees C. to about 550 degrees C.

Afterwards, a metal chloride-containing etching gas is applied on the metal oxide layer **147** to remove the metal oxide layer **147**, and the metal chloride-containing etching gas is applied on the metal layer **116'** to remove the native oxide layer **115** of the metal layer **116'** so as to form a

recessed metal layer **116''**, as shown in FIGS. **1H** and **2H**, in accordance with some embodiments. In some embodiments, the metal chloride-containing etching gas removes the metal oxide layer **147** and exposes top surfaces of the metal-semiconductor compound layers **145** and the top surface of the recessed metal layer **116''**, as shown in FIGS. **1H** and **2H**. In some embodiments, a top surface of the recessed metal layer **116''** is below a top surface of the high-k dielectric layer **114** as shown in FIGS. **1H** and **2H**. In some embodiments, the metal chloride-containing etching gas is made of $TaCl_5$. In some embodiments, the metal chloride-containing etching gas is made of $MoCl_5$, WCl_5 , BCl_3 , or a combination thereof.

In some embodiments, a dry etch process **138** is used to provide the metal chloride-containing etching gas to remove the metal oxide layer **147** and the native oxide layer **115** as shown in FIGS. **1H** and **2H**. In some embodiments, the operation temperature is in a range from about 300 degrees C. to about 500 degrees C. In some embodiments, the operation pressure is in a range from about 5 torrs to about 15 torrs. The removal of the metal oxide layer **147** and the native oxide layer **115** involves a chemical reaction between the metal chloride-containing etching gas and the oxides of the metal oxide layer **147** and the native oxide layer **115** at an elevated temperature, and this chemical reaction produces oxygen-containing gaseous products that can be removed, in accordance with some embodiments. Accordingly, such reaction mechanism can help the structure underneath the metal oxide layer **147** and the native oxide layer **115** to remain intact (e.g. the dimensions and the morphology of the structure including the metal-semiconductor compound layers **145** and the recessed metal layer **116''** remaining unchanged) after the removal of the metal oxide layer **147** and the native oxide layer **115**. In addition, because the metal nitride layer **141** and the remaining portions **143**, **144** and **146** of the metal material **140** are removed by oxidation and dry etch, the space for each of the subsequently formed metal contact features is enlarged, and resistance between the metal-semiconductor compound layer **145** and a subsequently formed metal contact feature can be reduced.

Afterwards, a metal contact feature **150** is formed over the top surface of the recessed metal layer **116''**, and metal contact features **152** are formed over the top surfaces of the metal-semiconductor compound layers **145**, as shown in FIGS. **1I** and **2I**, in accordance with some embodiments. In some embodiments, each of the metal contact features **150** and **152** is made of ruthenium. In some embodiments, each of the metal contact features **150** and **152** is made of cobalt, molybdenum, one or more other applicable materials, or a combination thereof. In some embodiments, the metal contact features **150** and **152** are selectively grown on the recessed metal layer **116''** and the metal-semiconductor compound layer **145**, respectively, so that the metal contact features **150** and **152** may have substantially seamless structure, and contact resistance can be reduced.

In some embodiments, a CVD process is used to form the metal contact features **150** and **152**. In some embodiments, the operation temperature is in a range from about 100 degrees C. to about 200 degrees C. In some embodiments, the operation pressure is in a range from about 5 torrs to about 15 torrs. In some embodiments, the reaction gas used for forming the metal contact features **150** and **152** includes $Ru_3(CO)_{12}$, $C_5HS(CO)_2Co$, $Co_2(CO)_8$, $Mo(CO)_6$, one or more other applicable precursors, or a combination thereof. In some embodiments, H_2 and/or Ar are also used. The thickness of the metal contact features **150** and **152** may be

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in a range from about 300 Å to about 1500 Å, which may be controlled by adjusting the operation time.

Afterwards, in some embodiments, a planarization process (not shown) is used to provide the metal contact features **150** and **152** with substantially planar top surfaces as shown in FIGS. 1I and 2I. The planarization process may include a chemical mechanical polishing (CMP) process, a dry polishing process, an etching process, a grinding process, one or more other applicable processes, or a combination thereof. In some embodiments, the top surfaces of the metal contact features **150** and **152** are substantially coplanar with the top surface of the dielectric layer **118**, as shown in FIGS. 1I and 2I.

In some other embodiments, an atomic layer deposition (ALD) process is used to form the metal contact features **150** and **152**. In some embodiments, the operation temperature is in a range from about 300 degrees C. to about 550 degrees C. In some embodiments, the operation pressure is in a range from about 15 torrs to about 40 torrs. In some embodiments, the reaction gas used for forming the metal contact features **150** and **152** includes a metal chloride-containing gas. In some embodiments, the reaction gas used for forming the metal contact features **150** and **152** includes RuCl_3 , MoCl_5 , one or more other applicable precursors, or a combination thereof. The metal chloride-containing gas used as the reaction gas in the ALD process may also act as an etching gas to clean the slightly oxidized surfaces which the metal contact features **150** and **152** are selectively grown on, followed by continuing to selectively form the metal contact features **150** and **152**. Accordingly, it can help to reduce resistance between the metal contact features **150** and **152** and the device elements (e.g. the conductive features including the metal gate stack **117** and the epitaxial structure **106**) thereunder. In some embodiments, H_2 and/or Ar are also used. The thickness of each of the metal contact features **150** and **152** may be in a range from about 300 Å to about 1500 Å, which may be controlled by adjusting the cycle number of the ALD process.

In some embodiments, the reaction gas used for forming the metal contact features **150** and **152** is substantially fluorine free, so that the device elements thereunder can be prevented from being negatively affected by fluorine.

In some embodiments, the application of the metal chloride-containing etching gas to remove the metal oxide layer **147** and the native oxide layer **115** as illustrated in FIGS. 1H and 2H and the formation of the metal contact features **150** and **152** as illustrated in FIGS. 1I and 2I are in-situ performed using the same process tool, e.g. the in-situ cluster tool **600** as shown in FIG. 6 in accordance with some embodiments, details of which will be discussed hereinafter. Since the above processes may be in-situ performed in the same process tool, the recessed metal layer **116**" and the metal-semiconductor compound layer **145** can be prevented from oxidized or polluted due to transferring between different process tools. Therefore, the interface between the metal contact feature **150** and the recessed metal layer **116**" and the interface between the metal contact feature **152** and the metal-semiconductor compound layer **145** can be substantially free from oxidation. Accordingly, the semiconductor structure can be prevented from polluted or damaged. In addition, the electrical connection between the metal contact features **150** and **152** and the conductive features including the metal gate stack **117** and the epitaxial structure **106** can be greatly improved.

In the embodiments illustrated in FIGS. 1A to 1I and FIGS. 2A to 2I, one metal contact structure including one metal contact feature **150** and two metal contact structures

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each including one metal contact feature **152** are formed on the top surface of the recessed metal layer **116**" and the metal-semiconductor compound layers **145**, respectively, leaving at least one metal layer **116** covered by the dielectric layer **118**. However, many variations and/or modifications can be made to embodiments of the disclosure. In some embodiments, the numbers of the metal contact feature(s) **150**, the metal contact feature(s) **152**, and the metal layer(s) **116** left un-connected to any metal contact features **150** and **152** may vary according to different needs.

Many variations and/or modifications can be made to embodiments of the disclosure. FIGS. 3A-3B are cross-sectional views of various stages of a process for forming a semiconductor structure **200**, in accordance with some embodiments of the disclosure. In some embodiments, a metal-containing precursor is applied on the recessed metal layer **116**" to selectively form a metal-containing element **120** on the recessed metal layer **116**", and the metal-containing precursor is applied on the metal-semiconductor compound layer **145** to form a metal-containing element **122** on each of the metal-semiconductor compound layers **145**, as shown in FIG. 3A. The metal-containing precursor may act as a selective metal growth precursor to selectively form the metal-containing elements **120** and **122** on the recessed metal layer **116**" and the metal-semiconductor compound layer **145**, respectively. The metal-containing elements **120** and **122** may act as seed layers for the metal contact features **150** and **152** to be formed thereon subsequently.

In some embodiments, the metal-containing precursor includes a tungsten-containing precursor. The tungsten-containing precursor may include WCl_5 , and the metal-containing elements **120** and **122** may be made of tungsten. In some embodiments, the metal-containing precursor is substantially fluorine free. Because the metal-containing precursor is substantially fluorine free, the device elements thereunder can be prevented from being negatively affected by fluorine. In some embodiments, the metal-containing precursor is selectively applied on the recessed metal layer **116**" and the metal-semiconductor compound layer **145** using an atomic layer deposition (ALD) process. In some embodiments, each of the metal-containing elements **120** and **122** has a thickness that is in a range from about 2 nm to about 4 nm, so that resistance between the metal-semiconductor compound layer **145** and the metal contact features **152** to be formed thereon can be reduced, and resistance between the recessed metal layer **116**" and the metal contact feature **150** to be formed thereon can be reduced.

In some embodiments, the operation temperature of the ALD process is in a range from about 300 degrees C. to about 550 degrees C. In some embodiments, the operation pressure of the ALD process may be in a range from about 15 torrs to about 40 torrs. In some embodiments, each of the metal-containing elements **120** and **122** may have a thickness that is in a range from about 10 Å to about 50 Å. The cycle number of the ALD process may be adjusted to control the thicknesses of the metal-containing elements **120** and **122**.

In some embodiments, the metal-containing precursor includes TaCl_5 , MoCl_5 , or a combination thereof. In these cases, the metal-containing elements **120** and **122** may be made of Ta or Mo.

In some embodiments, the top surfaces of the recessed metal layer **116**" and the metal-semiconductor compound layer **145** may be slightly oxidized to form native oxides on the top surfaces, and the metal-containing precursor may also act as an etching gas to remove the native oxides, followed by continuing to selectively form the metal-con-

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aining elements **120** and **122** on the recessed metal layer **116'** and the metal-semiconductor compound layer **145**, respectively. Thus, it can help to reduce resistance between the metal contact features **150** and **152** and the device elements (e.g. the conductive features including the metal gate stack **117** and the epitaxial structure **106**) thereunder. In some embodiments, the metal-containing element **120** is directly contacting the top surface of the recessed metal layer **116'** of the metal gate stack **117**, and each of the metal-containing elements **122** is directly contacting each of the metal-semiconductor compound layers **145**, as shown in FIG. 3A.

Afterwards, the metal contact feature **150** is selectively formed on the metal-containing element **120**, and the metal contact features **152** are selectively formed on the metal-containing elements **122** as shown in FIG. 3B, in accordance with some embodiments.

In some embodiments, the application of the metal chloride-containing etching gas to remove the metal oxide layer **147** and the native oxide layer **115** as illustrated in FIGS. 1H and 2H, the formation of the metal-containing elements **120** and **122** as illustrated in FIGS. 3A and 3B, and the formation of the metal contact features **150** and **152** as illustrated in FIGS. 1I and 2I are in-situ performed using the same process tool, e.g. the in-situ cluster tool **600** as shown in FIG. 6 in accordance with some embodiments, details of which will be discussed hereinafter. Similarly, since the above processes may be in-situ performed in the same process tool, the recessed metal layer **116'**, the metal-semiconductor compound layer **145**, and the metal-containing elements **120** and **122** can be prevented from oxidized or polluted due to transferring between different process tools. Therefore, the interfaces between the metal-containing elements **120** and **122** and the metal contact features **150** and **152**, the interface between the metal-containing element **120** and the recessed metal layer **116'**, and the interface between the metal-containing elements **122** and the metal-semiconductor compound layers **145** can be substantially free from oxidation. Thus, the semiconductor structure is prevented from polluted or damaged. In addition, the electrical connection between the metal contact structures including the metal contact features **150** and **152** and the metal-containing elements **120** and **122** and the conductive features including the metal gate stack **117** and the epitaxial structure **106** can be greatly improved.

In some embodiments, the metal contact structures include the metal contact features **150** and **152** made of ruthenium and/or molybdenum and the metal-containing elements **120** and **122** made of tungsten. In some embodiments, the metal contact structures include the metal contact features **150** and **152** made of ruthenium and the metal-containing elements **120** and **122** made of molybdenum.

FIGS. 4A-4E are cross-sectional views of various stages of a process for forming a semiconductor structure, in accordance with some embodiments of the disclosure. As shown in FIG. 4A, a structure the same as or similar to that shown in FIG. 2C is formed, in accordance with some embodiments.

Afterwards, a metal chloride-containing etching gas that is the same as or similar to the metal chloride-containing etching gas illustrated in FIGS. 1H and 2H is applied on the metal layer **116'** to remove the native oxide layer **115** so as to form a recessed metal layer **116'**, as shown in FIG. 4B, in accordance with some embodiments. In some embodiments, a dry etch process **148** that is the same as or similar to the dry etch process **138** illustrated in FIGS. 1H and 2H is used to provide the metal chloride-containing etching gas

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to remove the native oxide layer **115** as shown in FIG. 4B. In some embodiments, applying the metal chloride-containing etching gas on the metal layer **116'** is prior to forming the metal-containing layer over the dielectric layer **118**.

Afterwards, a metal material is deposited over the dielectric layer **118**, the top surfaces of the epitaxial structures **106**, and the recessed metal layer **116'**, and a portion of the metal material is nitride to form a metal nitride layer **441** over remaining portions (e.g. remaining portions **443**, **444**, and **446**) of the metal material, as shown in FIG. 4C, in accordance with some embodiments. The material and the formation method of the metal-containing layer including the metal nitride layer **441** and the remaining portions **443**, **444**, and **446** of the metal material may be the same as or similar to those of the metal-containing layer including the metal nitride layer **141** and the remaining portions **143**, **144**, and **146** of the metal material **140** illustrated in FIGS. 1D to 1E and FIGS. 2D to 2E.

Afterwards, the epitaxial structures **106**, the metal nitride layer **441**, and the remaining portions **444** of the metal material are heated to transform a first portion of the metal-containing layer contacting the top surfaces of the epitaxial structures **106** into the metal-semiconductor compound layers **445** as shown in FIG. 4C, in accordance with some embodiments. The material and the formation method of the metal-semiconductor compound layer **445** may be the same as or similar to those of the metal-semiconductor compound layer **145** illustrated in FIGS. 1F and 2F.

In some embodiments, the remaining portion **444** of the metal material is not fully transformed into the metal-semiconductor compound layer **445**, and a residual part (not shown) of the remaining portion **444** of the metal material remains untransformed between the metal-semiconductor compound layer **445** and the metal nitride layer **441**. In some embodiments, the residual part of the remaining portion **444** is made of a material that is the same as or similar to that of the remaining portion **444** of the metal material. In some embodiments, the residual part of the remaining portion **444** has a thickness that is in a range from about 1 nm to about 2 nm. In some embodiments, the residual part of the remaining portion **444** may be in form of a thin intervening layer located between the metal-semiconductor compound layer **445** and the metal nitride layer **441**.

Afterwards, the metal-containing layer is oxidized to transform a second portion of the metal-containing layer over the metal-semiconductor compound layers **445**, the recessed metal layer **116'**, and the dielectric layer **118** into a metal oxide layer **447** as shown in FIG. 4D, in accordance with some embodiments. The material and the formation method of the metal oxide layer **447** may be the same as or similar to those of the metal oxide layer **147** illustrated in FIGS. 1G and 2G.

After the metal-containing layer is oxidized, at least a third portion of the metal-containing layer may remain un-oxidized. Metal-containing portions **449** are formed between the metal-semiconductor compound layers **445** and the metal oxide layer **447**, and a metal-containing portion **451** is formed on the recessed metal layer **116'**, as shown in FIG. 4D, in accordance with some embodiments. In some embodiments, the metal-containing portions **449** and **451** are made of the material of the metal nitride layer **441**, the material of the remaining portions **443**, **444**, and **446** of the metal material, or a combination thereof. In some embodiments, the metal-containing portions **449** and **451** include the residual part of the remaining portion **444** of the metal material, so that the resistance between the epitaxial structure **106** and a metal contact feature to be formed over the

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metal-containing portion **449** and the metal-semiconductor compound layer **445** on the epitaxial structure **106** can be reduced, and the resistance between the recessed metal layer **116'** and a metal contact feature to be formed over the metal-containing portion **451** can be reduced.

In some embodiments, a plasma oxidation process is used to form the metal oxide layer **447** and the remaining metal-containing portions **449** and **451** as shown in FIG. **4D**. The operation temperature of the plasma oxidation process may be in a range from about 160 degrees C. to about 250 degrees C. The flow rate of oxygen gas may be in a range from about 2000 sccm to about 6000 sccm. Oxygen (O₂) plasma may be applied on the metal-containing layer. The O₂ plasma oxidation may be conducted with a source power that is in a range from about 100 W to about 1500 W and optionally with a bias power that is in a range from about 0 W to about 10 W. In some embodiments, argon gas is also used.

Afterwards, the metal oxide layer **447** is removed, and metal contact features **450** and **452** are selectively formed on the metal-containing portions **451** and **449**, respectively, as shown in FIG. **4E**, in accordance with some embodiments. The process used for removing the metal oxide layer **447** may be the same as or similar to that used for removing the metal oxide layer **147** as illustrated in FIGS. **1H** and **2H**. The material and the formation method of the metal contact features **450** and **452** may be the same as or similar to those of the metal contact features **150** and **152** illustrated in FIGS. **1I** and **2I**. In some embodiments, the metal contact structure including one metal-containing layer **450** and one metal-containing portion **451** is directly contacting the top surface of the recessed metal layer **116"**, and the metal contact structure including one metal-containing layer **452** and one metal-containing portion **449** is directly contacting the metal-semiconductor compound layer **445**, as shown in FIG. **4E**.

Many variations and/or modifications can be made to embodiments of the disclosure. In some embodiments, metal-containing elements **420** and **422** are selectively formed on the recessed metal layer **116"** and the metal-semiconductor compound layer **145**, respectively. FIG. **5** is a cross-sectional view of a semiconductor structure, in accordance with some embodiments of the disclosure. In some embodiments, the metal contact features **450** and **452** are selectively formed on the metal-containing elements **420** and **422**, respectively, as shown in FIG. **5**. The material and the formation method of the metal-containing elements **420** and **422** may be the same as or similar to those of the metal-containing elements **120** and **122** illustrated in FIGS. **3A** and **4A**.

FIG. **6** shows an in-situ cluster tool **600** used in a process for forming a semiconductor structure, in accordance with some embodiments of the disclosure. The in-situ cluster tool **600** includes process chambers **604**, **606**, **608**, **610**, **612**, **614**, **616**, **626**, and **628**, as shown in FIG. **6**, in accordance with some embodiments. Each of these chambers may be used for performing a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, a physical vapor deposition (PVD) process, or a dry etch process. Moreover, in some embodiments, the in-situ cluster tool **600** also includes moving mechanisms **630** and **632** for moving a treated or untreated intermediate semiconductor structure from one chamber to another chamber, as shown in FIG. **6**. In some embodiments, the in-situ cluster tool **600** also includes buffer chambers **634** and **636** for placing a treated or untreated intermediate semiconductor structure, as shown in FIG. **6**. In addition, in some embodiments, the in-situ cluster tool **600** further includes loading chambers **622** and

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624 connected to a buffer region **620**, and the in-situ cluster tool **600** also includes load docks **602** for connecting to other process tools, as shown in FIG. **6**.

In some embodiments, the application of the metal chloride-containing etching gas to remove the metal oxide layer **147** and the native oxide layer **115** as illustrated in FIGS. **1H** and **2H** is performed in one of the process chambers **604** and **606** in the in-situ cluster tool **600**, as shown in FIG. **6**. In some embodiments, the formation of the metal contact features **150** and **152** as illustrated in FIGS. **1I** and **2I** is performed in one of the process chambers **608**, **610** and **616** in the in-situ cluster tool **600** by chemical vapor deposition, as shown in FIG. **6**. In some embodiments, the above processes as illustrated in FIGS. **1H**, **2H**, **1I** and **2I** are in-situ performed in the same in-situ cluster tool **600** without breaking vacuum, and therefore, the recessed metal layer **116"** and the metal-semiconductor compound layer **145** can be prevented from oxidized or polluted due to transferring between different process tools.

In some embodiments, the formation of the metal-containing elements **120** and **122** as illustrated in FIGS. **3A** and **3B** is performed in one of the process chambers **612** and **614** in the in-situ cluster tool **600** by atomic layer deposition, as shown in FIG. **6**. Similarly, in some embodiments, the above processes as illustrated in FIGS. **1H**, **2H**, **3A**, **3B**, **1I** and **2I** are in-situ performed in the same in-situ cluster tool **600** without breaking vacuum, and therefore, the recessed metal layer **116"**, the metal-semiconductor compound layer **145**, and the metal-containing elements **120** and **122** can be prevented from oxidized or polluted due to transferring between different process tools.

As described previously, a portion of the metal material **140** is nitrified to form the metal nitride layer **141** over the remaining portions **143**, **144** and **146** of the metal material **140**. The metal nitride layer **141** may be used for protecting the remaining portions **143**, **144** and **146** of the metal material **140** from being oxidized or contamination, so that resistance between the metal contact feature **152** and the metal-semiconductor compound layer **145** formed from the remaining portion **144** can be reduced. In addition, the metal oxide layer **147** and the native oxide layer **115** are removed by the metal chloride-containing etching gas which results in oxygen-containing gaseous products that can be removed, thus such reaction mechanism can help the structure underneath the metal oxide layer **147** and the native oxide layer **115** to remain intact.

Moreover, the reaction gas used for forming the metal contact features **150** and **152** is substantially fluorine free, so that the device elements thereunder can be prevented from being negatively affected by fluorine. Furthermore, applying the metal chloride-containing etching gas to remove the metal oxide layer **147** and the native oxide layer **115**, forming the metal-containing elements **120** and **122**, and forming the metal contact features **150** and **152** are in-situ performed using the same process tool. Therefore, the semiconductor structure can be prevented from polluted or damaged, and the electrical connection between the metal contact features **150** and **152**, the metal-containing elements **120** and **122**, and the conductive features including the metal gate stack **117** and the epitaxial structure **106** can be greatly improved.

Embodiments for forming a semiconductor structure are provided. The method for forming the semiconductor structure may include transforming a portion of a metal-containing layer over a conductive feature into a metal-semiconductor compound layer, and removing another portion of the metal-containing layer by oxidation and dry etch, such that

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the space for a subsequently formed metal contact feature can be enlarged, and resistance between the metal-semiconductor compound layer and the metal contact feature can be reduced. In addition, the metal contact feature may be selectively grown on the metal-semiconductor compound layer, thus the electrical connection between the metal contact feature and the metal-semiconductor compound layer can be greatly improved.

In accordance with some embodiments, a method for forming a semiconductor structure is provided. The method includes forming an epitaxial structure over a semiconductor substrate. The method also includes forming a dielectric layer covering the epitaxial structure. The method further includes forming an opening in the dielectric layer to expose a top surface of the epitaxial structure. In addition, the method includes forming a metal-containing layer over the dielectric layer and the top surface of the epitaxial structure. The method includes heating the epitaxial structure and the metal-containing layer to transform a first portion of the metal-containing layer contacting the top surface of the epitaxial structure into a metal-semiconductor compound layer. The method also includes oxidizing the metal-containing layer to transform a second portion of the metal-containing layer over the metal-semiconductor compound layer and the dielectric layer into a metal oxide layer. The method further includes applying a metal chloride-containing etching gas on the metal oxide layer to remove the metal oxide layer. In addition, the method includes forming a metal contact feature over the top surface of the metal-semiconductor compound layer.

In accordance with some embodiments, a method for forming a semiconductor structure is provided. The method includes forming a metal layer and an epitaxial structure over a semiconductor substrate. The method also includes forming a dielectric layer covering the metal layer and the epitaxial structure. The method further includes forming a first opening and a second opening in the dielectric layer to expose the metal layer and the epitaxial structure respectively. In addition, the method includes applying a metal chloride-containing etching gas on the metal layer to remove a native oxide layer of the exposed metal layer so as to form a recessed metal layer. The method further includes forming a metal contact feature over a top surface of the recessed metal layer.

In accordance with some embodiments, a semiconductor structure is provided. The semiconductor structure includes an epitaxial structure over the semiconductor substrate. The semiconductor structure also includes a conductive feature over the semiconductor substrate. The conductive feature includes a high-k dielectric layer and a metal layer on the high-k dielectric layer, and a top surface of the metal layer is below a top surface of the high-k dielectric layer. The semiconductor structure further includes a metal-semiconductor compound layer formed on the epitaxial structure. In addition, the semiconductor structure includes a first metal contact structure formed on the top surface of the metal layer of the conductive feature. The semiconductor structure further includes a second metal contact structure formed on the metal-semiconductor compound layer.

In accordance with some embodiments, a semiconductor structure is provided. The semiconductor structure includes an epitaxial structure over a substrate and a metal layer of a gate stack adjacent to the epitaxial structure over the substrate. The semiconductor structure also includes a metal-semiconductor compound layer on the epitaxial structure. The semiconductor structure further includes a first metal-containing element on the metal layer of the gate stack and

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a second metal-containing element on the metal-semiconductor compound layer. A top surface of the metal layer of the gate stack is higher than a top surface of the second metal-containing element. In addition, the semiconductor structure includes a first metal contact feature over the first metal-containing element and a second metal contact feature over the second metal-containing element.

In accordance with some embodiments, a semiconductor structure is provided. The semiconductor structure includes a source/drain structure over a semiconductor substrate and a gate stack adjacent to the source/drain structure over the semiconductor substrate. The semiconductor structure also includes a metal-semiconductor compound layer on the source/drain structure. The semiconductor structure further includes a first metal-containing portion on a metal layer of the gate stack and a second metal-containing portion on the metal-semiconductor compound layer. In addition, the semiconductor structure includes a first metal-containing element on the first metal-containing portion and a second metal-containing element on the second metal-containing portion. The semiconductor structure further includes a first metal contact feature on the first metal-containing element and a second metal contact feature on the second metal-containing element.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor structure, comprising:

- an epitaxial structure over a semiconductor substrate;
- a conductive feature over the semiconductor substrate, wherein the conductive feature comprises a high-k dielectric layer and a metal layer on the high-k dielectric layer, wherein a top surface of the metal layer is below a top surface of the high-k dielectric layer;
- a metal-semiconductor compound layer formed on the epitaxial structure;
- a first metal contact structure formed over the top surface of the metal layer of the conductive feature;
- a second metal contact structure formed over the metal-semiconductor compound layer;
- a spacer between the epitaxial structure and the conductive feature, wherein the spacer is in direct contact with a lower portion of the second metal contact structure; and
- a dielectric layer on the spacer, wherein an interface between the dielectric layer and an upper portion of the second metal contact structure is substantially aligned with an interface between the spacer and a lower portion of the second metal contact structure.

2. The semiconductor structure as claimed in claim 1, wherein the first metal contact structure and the second metal contact structure each comprises a metal contact feature, and the metal contact feature comprises ruthenium, cobalt, molybdenum, or a combination thereof.

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3. The semiconductor structure as claimed in claim 1, wherein the first metal contact structure is in direct contact with a sidewall of the high-k dielectric layer of the conductive feature.

4. The semiconductor structure as claimed in claim 1, wherein the first metal contact structure is separated from the high-k dielectric layer of the conductive feature.

5. The semiconductor structure as claimed in claim 1, wherein a bottom surface of first metal contact structure is embedded in the conductive feature.

6. The semiconductor structure as claimed in claim 1, wherein a bottom surface of the second metal contact structure is lower than a top surface of the spacer.

7. The semiconductor structure as claimed in claim 1, wherein an interface between the dielectric layer and the spacer is higher than a bottom surface of the second metal contact structure.

8. The semiconductor structure as claimed in claim 1, wherein the upper portion of the second metal contact structure protrudes from the spacer, and the upper portion of the second metal contact structure and the lower portion of the second metal contact structure have same lateral width.

9. A semiconductor structure, comprising:

an epitaxial structure over a substrate;

a metal layer of a gate stack adjacent to the epitaxial structure over the substrate;

a metal-semiconductor compound layer on the epitaxial structure;

a first metal-containing element over the metal layer of the gate stack;

a second metal-containing element over the metal-semiconductor compound layer, wherein a bottom surface of the first metal-containing element is higher than a top surface of the second metal-containing element;

a first metal contact feature over the first metal-containing element;

a second metal contact feature over the second metal-containing element; and

a spacer on sidewalls of the gate stack, wherein the top surface of the second metal-containing element is lower than a top surface of the spacer, wherein the spacer includes a first spacer element adjacent to the sidewalls of the gate stack and a second spacer element adjacent to the first spacer element, wherein the second spacer element is in direct contact with a lower portion of the second metal contact feature.

10. The semiconductor structure as claimed in claim 9, wherein the second spacer element is made of a material different from that of the first spacer element.

11. The semiconductor structure as claimed in claim 9, wherein the second spacer element is in direct contact with the metal-semiconductor compound layer.

12. The semiconductor structure as claimed in claim 9, further comprising:

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a dielectric layer surrounding the first metal contact feature and the second metal contact feature over the epitaxial structure and the gate stack, wherein the first metal contact feature is separated from the second metal contact feature by the dielectric layer.

13. The semiconductor structure as claimed in claim 9, wherein materials of the first metal-containing element and the first metal contact feature are different.

14. A semiconductor structure, comprising:

a source/drain structure over a semiconductor substrate;

a gate stack adjacent to the source/drain structure over the semiconductor substrate;

a metal-semiconductor compound layer on the source/drain structure;

a first metal-containing portion on a metal layer of the gate stack;

a second metal-containing portion on the metal-semiconductor compound layer;

a first metal-containing element on the first metal-containing portion;

a second metal-containing element on the second metal-containing portion;

a first metal contact feature on the first metal-containing element;

a second metal contact feature on the second metal-containing element; and

a spacer in direct contact with a sidewall of the second metal-containing element and a sidewall of the second metal contact feature, wherein a sidewall of the second metal-containing portion and a sidewall of the metal-semiconductor compound layer are contacted by the spacer.

15. The semiconductor structure as claimed in claim 14, wherein an interface between the first metal-containing element and the first metal-containing portion is lower than a top surface of a gate dielectric layer of the gate stack.

16. The semiconductor structure as claimed in claim 14, wherein an interface between the first metal-containing element and the first metal-containing portion is higher than the interface between the second metal-containing element and the second metal-containing portion.

17. The semiconductor structure as claimed in claim 14, wherein a top surface of the second metal-containing element is lower than the top surface of the spacer.

18. The semiconductor structure as claimed in claim 14, wherein a bottom surface of the second metal-containing portion is higher than a bottom surface of the spacer.

19. The semiconductor structure as claimed in claim 14, wherein a top surface of the second metal-containing portion is higher than a bottom surface of the spacer.

20. The semiconductor structure as claimed in claim 14, wherein a bottom surface of the metal-semiconductor compound layer is higher than a bottom surface of the spacer.

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